STA260: Probability and Statistics II

Personal Reference Notebook

University of Toronto Mississauga — Fall 2025

Textbook: D. D. Wackerly, W. Mendenhall & R. L. Scheaffer, *Mathematical Statistics with Applications*, 7th ed.

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A personal course notebook containing my own solutions to lectures, tutorials, assignments, and exams from STA260 Fall 2025. Compiled as a self-contained reference for probability and mathematical statistics.

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Notation and Conventions

Symbol	Meaning
X, Y, Z	Random variables
x, y, z	Realizations / observed values
$\theta, \mu, \sigma^2, \lambda, \beta$	Parameters
$\hat{\theta}, \hat{\mu}, \hat{p}$	Estimators
$\mathbb{E}[X]$	Expected value of X
$\mathrm{Var}(X)$	Variance of X
$\mathrm{Cov}(X, Y)$	Covariance of X and Y
$\mathbb{P}(A)$	Probability of event A
$M_X(t)$	Moment generating function of X
$f(x)$, $f(x \mid \theta)$	PDF or PMF
$F(x)$	CDF
$L(\theta \mid \boldsymbol{x})$	Likelihood function
$\ell(\theta)$	Log-likelihood function
$\mathcal{I}(\theta)$	Fisher information (single observation)
$n\,\mathcal{I}(\theta)$	Fisher information (n observations)
$\mathbf{1}_A$	Indicator function of event A
$X \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$	iid draws from Normal
$X_{(1)}, X_{(n)}$	Minimum, maximum order statistics
$\mathrm{d}x$	Differential

Environment Colour Guide:

- Definitions
- Theorems and Results
- Key Results and Formulas to Memorize
- Worked Examples
- My Solved Problems (from notes, tests, assignments)
- Strategy Boxes
- Warnings and Common Mistakes

Chapter 1

Review of Probability Foundations

This chapter reviews the prerequisite probability from STA256 that forms the foundation for the course.

1.1 Sample Spaces and Events

Definition 1.1 (Sample Space).

The **sample space** S of an experiment is the set of all possible outcomes. An **event** A is any subset $A \subseteq S$.

1.2 Axioms of Probability

Definition 1.2 (Kolmogorov Axioms).

A probability function $\mathbb{P}$ on a sample space S satisfies:

1. $\mathbb{P}(A) \geq 0$ for every event A.
2. $\mathbb{P}(S) = 1$.
3. If $A_1, A_2, \ldots$ are mutually exclusive, then $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

1.3 Conditional Probability and Independence

Definition 1.3 (Conditional Probability).

If $\mathbb{P}(B) > 0$, the conditional probability of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Definition 1.4 (Independence).

Events A and B are **independent** if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$.

1.4 Bayes' Theorem and the Law of Total Probability

Theorem 1.1 (Law of Total Probability).

If $B_1, B_2, \ldots, B_k$ partition S with $\mathbb{P}(B_i) > 0$ for all i, then

$$\mathbb{P}(A) = \sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i).$$

Theorem 1.2 (Bayes' Theorem).

Under the same partition,

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\,\mathbb{P}(B_j)}{\sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\,\mathbb{P}(B_i)}.$$

1.5 Expectation and Variance

Definition 1.5 (Expectation and Variance).

For a random variable X:

$$\mathbb{E}[X] = \begin{cases} \sum_x x\, f(x) & \text{(discrete)} \\ \int_{-\infty}^{\infty} x\, f(x)\,\mathrm{d}x & \text{(continuous)} \end{cases}$$

$$\mathrm{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

Key Result 1.1 Properties of Expectation and Variance.

1. $\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b$
2. $\mathrm{Var}(aX + b) = a^2\,\mathrm{Var}(X)$
3. $\mathbb{E}[g(X) + h(X)] = \mathbb{E}[g(X)] + \mathbb{E}[h(X)]$
4. If X, Y independent: $\mathbb{E}[XY] = \mathbb{E}[X]\,\mathbb{E}[Y]$ and $\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y)$

Solved Problem 1.1 (Sample Proportion via CLT (Note Sep 14)).

Problem. Suppose $p = 0.02$, $n = 2000$. Find $\mathbb{P}(\hat{p} \geq 0.03)$.

Solution. By CLT: $\hat{p} \dot{\sim} \mathcal{N}\left(p,\ \frac{p(1-p)}{n}\right)$, so SE $= \sqrt{\frac{0.02(0.98)}{2000}} = 0.00313$.

$$Z = \frac{0.03 - 0.02}{0.00313} = 3.195.$$

$$\mathbb{P}(\hat{p} \geq 0.03) = \mathbb{P}(Z \geq 3.195) \approx \boxed{0.0007}.$$

Extremely unlikely to observe $\hat{p} \geq 0.03$ if the true proportion is 0.02.

Solved Problem 1.2 (Large n with Unknown σ (Note Sep 14)).

Problem. $s = 12$, $n = 400$. Find $\mathbb{P}(\bar{X} > \mu + 1)$.

Solution. When n is large, $s \approx \sigma$: $\mathrm{SE} = s/\sqrt{n} = 12/\sqrt{400} = 0.6$.

$$Z = \frac{1}{0.6} = 1.667, \qquad \mathbb{P}(\bar{X} > \mu + 1) = \mathbb{P}(Z > 1.667) \approx \boxed{0.048}.$$

Chapter 2

Common Discrete Distributions

2.1 Bernoulli Distribution

Definition 2.1 (Bernoulli Distribution).

$X \sim \text{Bernoulli}(p)$ if $X \in \{0, 1\}$ with PMF

$$f(x) = p^x(1-p)^{1-x}, \quad x \in \{0, 1\}.$$

$\mathbb{E}[X] = p, \quad \text{Var}(X) = p(1-p), \quad M_X(t) = (1-p) + pe^t.$

2.2 Binomial Distribution

Definition 2.2 (Binomial Distribution).

$X \sim \text{Bin}(n, p)$ if X counts successes in n independent Bernoulli trials:

$$f(x) = \binom{n}{x} p^x(1-p)^{n-x}, \quad x = 0, 1, \ldots, n.$$

$\mathbb{E}[X] = np, \quad \text{Var}(X) = np(1-p), \quad M_X(t) = [(1-p) + pe^t]^n.$

Key Result 2.1 Normal Approximation to Binomial.

If $np \geq 10$ and $n(1-p) \geq 10$, then $X \sim \text{Bin}(n, p)$ is approximately $\mathcal{N}(np, np(1-p))$.
Continuity correction: For integer boundaries, use half-integer adjustments:

$$\mathbb{P}(a \leq X \leq b) \approx \mathbb{P}\left(\frac{a - 0.5 - np}{\sqrt{np(1-p)}} \leq Z \leq \frac{b + 0.5 - np}{\sqrt{np(1-p)}}\right).$$

Example 2.1 (Normal Approximation to Binomial (Test 2, Fall 2023, Q1)).

Problem. Let $X \sim \text{Bin}(150, 0.10)$. Find $\mathbb{P}(15 \leq X \leq 20)$.

Solution. Compute $\mu = np = 15$, $\sigma^2 = np(1-p) = 13.5$, $\sigma = 3.674$.
Check: $np = 15 \geq 10$ ✓, $n(1-p) = 135 \geq 10$ ✓.

Apply continuity correction:

$$\mathbb{P}(15 \leq X \leq 20) \approx \mathbb{P}(14.5 \leq X \leq 20.5).$$

Standardize:

$$Z_L = \frac{14.5 - 15}{3.674} = -0.136, \qquad Z_U = \frac{20.5 - 15}{3.674} = 1.497.$$

$$\mathbb{P}(-0.136 \leq Z \leq 1.497) = \Phi(1.497) - \Phi(-0.136) \approx 0.9332 - 0.4443 = \boxed{0.489}.$$

2.3 Geometric Distribution

Definition 2.3 (Geometric Distribution).

$X \sim \text{Geom}(p)$ counts the number of trials until the first success:

$$f(x) = p(1-p)^{x-1}, \quad x = 1, 2, 3, \ldots$$

$\mathbb{E}[X] = 1/p, \quad \text{Var}(X) = (1-p)/p^2, \quad M_X(t) = \dfrac{pe^t}{1-(1-p)e^t}$ for $t < -\ln(1-p)$.

2.4 Negative Binomial Distribution

Definition 2.4 (Negative Binomial Distribution).

$X \sim \text{NegBin}(r, p)$ counts trials until the r-th success:

$$f(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, \ldots$$

$\mathbb{E}[X] = r/p, \quad \text{Var}(X) = r(1-p)/p^2.$

2.5 Hypergeometric Distribution

Definition 2.5 (Hypergeometric Distribution).

Drawing n items without replacement from N items (r successes, $N - r$ failures):

$$f(x) = \frac{\binom{r}{x}\binom{N-r}{n-x}}{\binom{N}{n}}, \quad x = \max(0, n-N+r), \ldots, \min(n, r).$$

$\mathbb{E}[X] = nr/N, \quad \text{Var}(X) = n \cdot \dfrac{r}{N} \cdot \dfrac{N-r}{N} \cdot \dfrac{N-n}{N-1}.$

2.6 Poisson Distribution

Definition 2.6 (Poisson Distribution).

$X \sim \text{Poisson}(\lambda)$ if

$$f(x) = \frac{\lambda^x e^{-\lambda}}{x!}, \quad x = 0, 1, 2, \ldots$$

$\mathbb{E}[X] = \lambda, \quad \text{Var}(X) = \lambda, \quad M_X(t) = e^{\lambda(e^t - 1)}.$

2.7 Summary Table

Distribution	$\mathbf{E}[X]$	$\mathbf{Var}(X)$	**MGF**
Bernoulli(p)	p	$p(1-p)$	$(1-p) + pe^t$
Bin(n, p)	np	$np(1-p)$	$[(1-p) + pe^t]^n$
Geom(p)	$1/p$	$(1-p)/p^2$	$pe^t/[1-(1-p)e^t]$
Poisson(λ)	λ	λ	$e^{\lambda(e^t-1)}$

Chapter 3

Common Continuous Distributions

3.1 Uniform Distribution

Definition 3.1 (Uniform Distribution).

$X \sim \text{Uniform}(a, b)$ if

$$f(x) = \frac{1}{b-a}, \quad a \le x \le b.$$

$\mathbb{E}[X] = \dfrac{a+b}{2}, \quad \text{Var}(X) = \dfrac{(b-a)^2}{12}, \quad F(x) = \dfrac{x-a}{b-a}.$

3.2 Exponential Distribution

Definition 3.2 (Exponential Distribution).

$X \sim \text{Exp}(\theta)$ with mean $\theta > 0$:

$$f(x) = \frac{1}{\theta} e^{-x/\theta}, \quad x > 0.$$

$\mathbb{E}[X] = \theta, \quad \text{Var}(X) = \theta^2, \quad F(x) = 1 - e^{-x/\theta}, \quad M_X(t) = (1-\theta t)^{-1}$ for $t < 1/\theta$.

Remark 3.1. *Some texts parametrize with rate $\lambda = 1/\theta$, giving $f(x) = \lambda e^{-\lambda x}$ and $\mathbb{E}[X] = 1/\lambda$.*

3.3 Gamma Distribution

Definition 3.3 (Gamma Distribution).

$X \sim \text{Gamma}(\alpha, \beta)$ if

$$f(x) = \frac{1}{\Gamma(\alpha)\,\beta^\alpha}\, x^{\alpha-1}\, e^{-x/\beta}, \quad x > 0.$$

$\mathbb{E}[X] = \alpha\beta, \quad \text{Var}(X) = \alpha\beta^2, \quad M_X(t) = (1-\beta t)^{-\alpha}$ for $t < 1/\beta$.
Special cases: $\text{Gamma}(1, \beta) = \text{Exp}(\beta); \quad \text{Gamma}(n/2, 2) = \chi^2(n)$.

Key Result 3.1 Gamma Function.

$\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x}\,\mathrm{d}x$, with $\Gamma(n) = (n-1)!$ for positive integers and $\Gamma(1/2) = \sqrt{\pi}$.

3.4 Normal Distribution

Definition 3.4 (Normal Distribution).

$X \sim \mathcal{N}(\mu, \sigma^2)$ if

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}.$$

$\mathbb{E}[X] = \mu, \quad \mathrm{Var}(X) = \sigma^2, \quad M_X(t) = \exp\left(\mu t + \frac{\sigma^2 t^2}{2}\right).$
Standardization: $Z = (X-\mu)/\sigma \sim \mathcal{N}(0,1)$.

Key Result 3.2 Sum of Independent Normals.

If $X_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ independently, then $\sum_{i=1}^n a_i X_i \sim \mathcal{N}(\sum a_i \mu_i,\ \sum a_i^2 \sigma_i^2)$.

3.5 Beta Distribution

Definition 3.5 (Beta Distribution).

$X \sim \mathrm{Beta}(\alpha, \beta)$ if

$$f(x) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}, \quad 0 < x < 1.$$

$\mathbb{E}[X] = \dfrac{\alpha}{\alpha+\beta}, \quad \mathrm{Var}(X) = \dfrac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$
Special case: $\mathrm{Beta}(\theta, 1)$ has PDF $f(x) = \theta\, x^{\theta-1}$ on $(0,1)$.

3.6 Chi-Squared Distribution

Definition 3.6 (Chi-Squared Distribution).

If $Z_1, \ldots, Z_n \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, then $W = \sum_{i=1}^n Z_i^2 \sim \chi^2(n)$, which is $\mathrm{Gamma}(n/2, 2)$.
$\mathbb{E}[W] = n, \quad \mathrm{Var}(W) = 2n.$

Theorem 3.1 (Moments of Chi-Squared (Tutorial 1, Q1)).

If $X \sim \chi^2(r)$ and $k > -r/2$, then

$$\mathbb{E}[X^k] = \frac{2^k\,\Gamma(r/2+k)}{\Gamma(r/2)}.$$

Solved Problem 3.1 (Proof of Chi-Squared Moments Formula (TUT 1, Q1)).

Problem. Prove the moments formula above.

Solution. $X \sim \chi^2(r) = \text{Gamma}(r/2, 2)$, so $f_X(x) = \dfrac{x^{r/2-1}e^{-x/2}}{2^{r/2}\,\Gamma(r/2)}$, $x > 0$.

$$\mathbb{E}[X^k] = \int_0^\infty x^k \cdot \frac{x^{r/2-1}e^{-x/2}}{2^{r/2}\Gamma(r/2)}\,\mathrm{d}x = \frac{1}{2^{r/2}\Gamma(r/2)}\int_0^\infty x^{r/2+k-1}e^{-x/2}\,\mathrm{d}x.$$

Substituting $u = x/2$ ($x = 2u,\ \mathrm{d}x = 2\,\mathrm{d}u$):

$$= \frac{1}{2^{r/2}\Gamma(r/2)}\int_0^\infty (2u)^{r/2+k-1}e^{-u}\cdot 2\,\mathrm{d}u = \frac{2^{r/2+k}}{2^{r/2}\Gamma(r/2)}\int_0^\infty u^{r/2+k-1}e^{-u}\,\mathrm{d}u.$$

The integral is $\Gamma(r/2+k)$ (valid when $r/2+k > 0$, i.e., $k > -r/2$):

$$\boxed{\mathbb{E}[X^k] = \frac{2^k\,\Gamma(r/2+k)}{\Gamma(r/2)}}.$$

Special cases: $k = 1$: $\mathbb{E}[X] = 2 \cdot \frac{\Gamma(r/2+1)}{\Gamma(r/2)} = 2 \cdot r/2 = r$. $k = 2$: $\mathbb{E}[X^2] = 4 \cdot \frac{\Gamma(r/2+2)}{\Gamma(r/2)} = r^2 + 2r$.

Key Result 3.3 Sampling Distribution of S^2.

If $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then:

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1), \qquad \text{where } S^2 = \frac{1}{n-1}\sum_{i=1}^n (Y_i - \bar{Y})^2.$$

Moreover, $\bar{Y}$ and S^2 are independent.

3.7 Summary Table

Distribution	$\mathbb{E}[X]$	$\mathrm{Var}(X)$	**MGF**
Uniform(a, b)	$(a+b)/2$	$(b-a)^2/12$	$(e^{tb} - e^{ta})/[t(b-a)]$
Exp(θ)	θ	θ^2	$(1-\theta t)^{-1}$
Gamma(α, β)	$\alpha\beta$	$\alpha\beta^2$	$(1-\beta t)^{-\alpha}$
$\mathcal{N}(\mu, \sigma^2)$	μ	σ^2	$e^{\mu t + \sigma^2 t^2/2}$
Beta(α, β)	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	—
$\chi^2(n)$	n	$2n$	$(1-2t)^{-n/2}$

Chapter 4

Moment Generating Functions

4.1 Definition and Motivation

Definition 4.1 (Moment Generating Function).

The **moment generating function** (MGF) of a random variable X is

$$M_X(t) = \mathbb{E}[e^{tX}],$$

provided this expectation exists in a neighbourhood of $t = 0$.

Theorem 4.1 (Moments from the MGF).

The k-th moment of X is obtained by evaluating the k-th derivative at zero:

$$\mathbb{E}[X^k] = M_X^{(k)}(0) = \frac{d^k}{dt^k} M_X(t)\bigg|_{t=0}.$$

In particular: $\mathbb{E}[X] = M_X'(0)$ and $\mathbb{E}[X^2] = M_X''(0)$.

4.2 Properties of MGFs

Key Result 4.1 Transformation Properties.

1. **Scaling:** If $Y = aX$, then $M_Y(t) = M_X(at)$.
2. **Translation:** If $Y = a + X$, then $M_Y(t) = e^{at} M_X(t)$.
3. **Independence:** If X_1, X_2 are independent and $Y = X_1 + X_2$, then $M_Y(t) = M_{X_1}(t) \cdot M_{X_2}(t)$.

4.3 Uniqueness Theorem

Theorem 4.2 (Uniqueness of the MGF).

If the MGF $M_X(t)$ exists in a neighbourhood of $t = 0$, it uniquely determines the distribution of X. That is, if $M_X(t) = M_Y(t)$ for all t in a neighbourhood of 0, then X and Y have the same distribution.

Remark 4.1. *This is the key property that makes the "MGF method" work: to identify a distribution, compute the MGF and match it to a known form.*

4.4 MGF Method for Sampling Distributions

Example 4.1 (Distribution of $\bar{X}$ via MGF (Lecture Notes)).

Problem. Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. Find the distribution of $\bar{X} = \frac{1}{n}\sum X_i$.

Solution. Write $\bar{X} = \frac{1}{n}\sum X_i$. By independence:

$$M_{\bar{X}}(t) = \mathbb{E}\left[e^{t\bar{X}}\right] = \mathbb{E}\left[\prod_{i=1}^{n} e^{tX_i/n}\right] = \prod_{i=1}^{n} M_{X_i}(t/n).$$

Each $X_i \sim \mathcal{N}(\mu, \sigma^2)$ has $M_{X_i}(s) = \exp(\mu s + \sigma^2 s^2/2)$. Substituting $s = t/n$:

$$M_{\bar{X}}(t) = \left[\exp\left(\frac{\mu t}{n} + \frac{\sigma^2 t^2}{2n^2}\right)\right]^n = \exp\left(\mu t + \frac{\sigma^2 t^2}{2n}\right).$$

This is the MGF of $\mathcal{N}(\mu, \sigma^2/n)$. By uniqueness, $\boxed{\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)}$.

4.5 Worked Examples

Example 4.2 (Sum of Independent Normals (Practice MGF, Q6)).

Problem. $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ independent. Find the distribution of $Y = X_1 + 3X_2$.

Solution.

$$M_Y(t) = M_{X_1}(t) \cdot M_{X_2}(3t) = e^{\mu_1 t + \sigma_1^2 t^2/2} \cdot e^{3\mu_2 t + 9\sigma_2^2 t^2/2} = e^{(\mu_1 + 3\mu_2)t + (\sigma_1^2 + 9\sigma_2^2)t^2/2}.$$

By uniqueness: $\boxed{Y \sim \mathcal{N}(\mu_1 + 3\mu_2,\ \sigma_1^2 + 9\sigma_2^2)}$.

Example 4.3 (MGF of a Double Exponential (Practice MGF, Q11)).

Problem. X has PDF $f(x) = \frac{1}{2}e^{-|x|}$ for $x \in \mathbb{R}$.
(a) Find k such that $f(x) = ke^{-|x|}$ is a valid PDF.
(b) Find $M_X(t)$.
(c) Find $\mathbb{E}[X]$ and $\mathrm{Var}(X)$.

Solution.
(a) $\int_{-\infty}^{\infty} ke^{-|x|}\,\mathrm{d}x = 2k\int_0^{\infty} e^{-x}\,\mathrm{d}x = 2k = 1$, so $\boxed{k = 1/2}$.
(b) For $|t| < 1$:

$$\begin{aligned} M_X(t) &= \tfrac{1}{2}\int_{-\infty}^{0} e^{(t+1)x}\,\mathrm{d}x + \tfrac{1}{2}\int_0^{\infty} e^{(t-1)x}\,\mathrm{d}x \\ &= \frac{1}{2}\cdot\frac{1}{t+1} + \frac{1}{2}\cdot\frac{1}{1-t} = \boxed{\frac{1}{1-t^2}}. \end{aligned}$$

(c) $M_X'(t) = 2t/(1-t^2)^2$, so $\mathbb{E}[X] = M_X'(0) = \boxed{0}$.
$M_X''(t) = (2+6t^2)/(1-t^2)^3$, so $\mathbb{E}[X^2] = M_X''(0) = 2$, giving $\mathrm{Var}(X) = 2 - 0 = \boxed{2}$.

Solved Problem 4.1 (Cumulant Generating Function (Practice MGF, Q12)).

Problem. Let $S(t) = \ln M_X(t)$ (the *cumulant generating function*). Show:
(a) $S(0) = 0$.
(b) $S'(0) = \mathbb{E}[X]$.
(c) $S''(0) = \mathrm{Var}(X)$.

Solution.
(a) $S(0) = \ln M_X(0) = \ln \mathbb{E}[e^0] = \ln 1 = \boxed{0}$.
(b) $S'(t) = M_X'(t)/M_X(t)$. At $t = 0$: $S'(0) = M_X'(0)/M_X(0) = \mathbb{E}[X]/1 = \boxed{\mathbb{E}[X]}$.
(c) By the quotient rule:

$$S''(t) = \frac{M_X''(t)M_X(t) - [M_X'(t)]^2}{[M_X(t)]^2}.$$

At $t = 0$: $S''(0) = \mathbb{E}[X^2] - [\mathbb{E}[X]]^2 = \boxed{\mathrm{Var}(X)}$.
The cumulant generating function is a useful alternative to the MGF: its first derivative at 0 gives the mean, and its second derivative gives the variance directly.

Example 4.4 (MGF of Linear Combination $Y = X_1 + 3X_2$ (Practice MGF, Q6)).

Problem. $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ independent. Find the distribution of $Y = X_1 + 3X_2$.

Solution. $M_Y(t) = M_{X_1}(t) \cdot M_{X_2}(3t) = e^{\mu_1 t + \sigma_1^2 t^2/2} \cdot e^{3\mu_2 t + 9\sigma_2^2 t^2/2}$.

$$M_Y(t) = \exp\left[(\mu_1 + 3\mu_2)t + \frac{(\sigma_1^2 + 9\sigma_2^2)t^2}{2}\right].$$

This is the MGF of $\mathcal{N}(\mu_1 + 3\mu_2,\ \sigma_1^2 + 9\sigma_2^2)$.
$\boxed{Y \sim \mathcal{N}(\mu_1 + 3\mu_2,\ \sigma_1^2 + 9\sigma_2^2)}$.

Chapter 5

Transformations of Random Variables

5.1 The CDF Method

CDF Method for Transformations To find the distribution of $Y = g(X)$:

1. Compute $F_Y(y) = \mathbb{P}(Y \leq y) = \mathbb{P}(g(X) \leq y)$.
2. Rewrite in terms of X: solve $g(X) \leq y$ for X.
3. Substitute the known CDF of X.
4. Differentiate: $f_Y(y) = F_Y'(y)$.

Example 5.1 (Transformation via CDF (Tutorial 1, Q2)).

Problem. Let $X \sim \text{Uniform}(0, 1)$. Find the CDF and PDF of $Y = -2 \ln X$.

Solution.

$$\begin{aligned} F_Y(y) &= \mathbb{P}(-2 \ln X \leq y) = \mathbb{P}(\ln X \geq -y/2) = \mathbb{P}(X \geq e^{-y/2}) \\ &= 1 - e^{-y/2}, \quad y > 0. \end{aligned}$$

Differentiate:

$$f_Y(y) = \frac{1}{2} e^{-y/2}, \quad y > 0.$$

This is the PDF of Exp(2), i.e., $\boxed{Y \sim \chi^2(2)}$.

5.2 The PDF Method (Change of Variables)

Theorem 5.1 (One-to-One Change of Variables).

If $Y = g(X)$ where g is a one-to-one differentiable function with inverse $X = g^{-1}(Y)$, then

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|.$$

5.3 The MGF Method

MGF Method for Transformations To find the distribution of $Y = g(X)$:

1. Compute $M_Y(t) = \mathbb{E}[e^{tY}] = \mathbb{E}[e^{tg(X)}]$.
2. Match the result to the MGF of a known distribution.
3. By uniqueness, Y has that distribution.

This is especially powerful for sums of independent random variables.

Example 5.2 (PDF Method — Jacobian Technique (Final Exam Fall 2025, Q1b)).

Problem. Let $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ and $Y_{(1)} = \min(Y_1, \ldots, Y_n)$. Use the Jacobian technique to find the distribution of $Q = 2nY_{(1)}/\theta$.

Solution. From the minimum-of-exponentials result, $Y_{(1)} \sim \text{Exp}(\theta/n)$ with PDF $f_{Y_{(1)}}(y) = (n/\theta)e^{-ny/\theta}$, $y > 0$.
Let $Q = 2nY_{(1)}/\theta$. Then $Y_{(1)} = Q\theta/(2n)$ and $|dY_{(1)}/dQ| = \theta/(2n)$.
By the change-of-variables formula:

$$f_Q(q) = f_{Y_{(1)}}\left(\frac{q\theta}{2n}\right) \cdot \frac{\theta}{2n} = \frac{n}{\theta}\, e^{-n \cdot q\theta/(2n\theta)} \cdot \frac{\theta}{2n} = \frac{1}{2} e^{-q/2}, \quad q > 0.$$

This is the PDF of $\text{Gamma}(1, 2) = \chi^2(2)$. Therefore $\boxed{Q = 2nY_{(1)}/\theta \sim \chi^2(2)}$.

Example 5.3 (CDF Method for Rayleigh (Summer 2025 Test 2, Q3)).

Problem. Let $X \sim \text{Exp}(\theta)$ and $Y = \sqrt{X}$. Find the PDF of Y.

Solution. For $y > 0$:

$$F_Y(y) = \mathbb{P}(\sqrt{X} \leq y) = \mathbb{P}(X \leq y^2) = 1 - e^{-y^2/\theta}.$$

Differentiate:

$$f_Y(y) = \frac{2y}{\theta}\, e^{-y^2/\theta}, \quad y > 0.$$

This is the Rayleigh distribution with parameter θ. Note $Y^2 \sim \text{Exp}(\theta)$.

5.4 Multivariable Transformations

Theorem 5.2 (Jacobian Method for Two Variables).

If $(U, V) = (g_1(X, Y), g_2(X, Y))$ is a one-to-one transformation with inverse $(X, Y) = (h_1(U, V), h_2(U, V))$, then

$$f_{U,V}(u, v) = f_{X,Y}(h_1(u, v), h_2(u, v))\,|J|,$$

where the Jacobian is $J = \det \begin{pmatrix} \partial x/\partial u & \partial x/\partial v \\ \partial y/\partial u & \partial y/\partial v \end{pmatrix}$.

Solved Problem 5.1 (Logistic Distribution via CDF (TT1 Review Session)).

Problem. Let Y have PDF $f(y; \theta) = \dfrac{e^{(y-\theta)}}{\left[1 + e^{(y-\theta)}\right]^2}$, $y \in \mathbb{R}$. Define $Q = Y - \theta$.
(a) Show Q is a pivot (distribution free of θ).
(b) Find $F_Q(a)$.

Solution.
(a) The PDF of Q is $f_Q(q) = \dfrac{e^q}{(1 + e^q)^2}$, which does not depend on θ. So Q is a pivot.
(b) $F_Q(a) = \int_{-\infty}^{a} \frac{e^q}{(1+e^q)^2}\,\mathrm{d}q$.
Let $u = 1 + e^q$, $\mathrm{d}u = e^q\,\mathrm{d}q$:

$$F_Q(a) = \int \frac{1}{u^2}\,\mathrm{d}u = -\frac{1}{u}\bigg|_{-\infty}^{a} = -\frac{1}{1 + e^a} + 1 = \boxed{\frac{e^a}{1 + e^a}}.$$

This is the standard logistic CDF.

Chapter 6

Joint Distributions and Marginals

6.1 Joint PMF and PDF

Definition 6.1 (Joint Distribution).

For continuous (X, Y), the **joint PDF** $f(x, y)$ satisfies $f(x, y) \geq 0$ and $\iint f(x, y)\,\mathrm{d}x\,\mathrm{d}y = 1$. For discrete (X, Y), the **joint PMF** $f(x, y) = \mathbb{P}(X = x, Y = y)$.

6.2 Marginal Distributions

Definition 6.2 (Marginal Distributions).

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\,\mathrm{d}y, \qquad f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\,\mathrm{d}x.$$

6.3 Independence of Random Variables

Definition 6.3 (Independence).

X and Y are independent if and only if $f(x, y) = f_X(x) \cdot f_Y(y)$ for all (x, y).

6.4 Covariance and Correlation

Definition 6.4 (Covariance and Correlation).

$$\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y], \qquad \mathrm{Corr}(X, Y) = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\,\mathrm{Var}(Y)}}.$$

Key Result 6.1 Variance of a Sum.

$$\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y).$$

If X, Y independent: $\mathrm{Cov}(X, Y) = 0$, so $\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}(X) + b^2\,\mathrm{Var}(Y)$.

Chapter 7

Conditional Distributions and Expectation

7.1 Conditional PDF and PMF

Definition 7.1 (Conditional Distribution).

The conditional PDF of Y given $X = x$ is

$$f(y \mid x) = \frac{f(x, y)}{f_X(x)}, \quad \text{provided } f_X(x) > 0.$$

7.2 Conditional Expectation

Definition 7.2 (Conditional Expectation).

$$\mathbb{E}[Y \mid X = x] = \int_{-\infty}^{\infty} y\, f(y \mid x)\, \mathrm{d}y.$$

7.3 Law of Total Expectation

Theorem 7.1 (Law of Total Expectation (Adam's Law)).

$$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid X]].$$

7.4 Law of Total Variance

Theorem 7.2 (Law of Total Variance (Eve's Law)).

$$\mathrm{Var}(Y) = \mathbb{E}[\mathrm{Var}(Y \mid X)] + \mathrm{Var}(\mathbb{E}[Y \mid X]).$$

Chapter 8

Functions of Multiple Random Variables

8.1 Distributions of Sums

Theorem 8.1 (Sum of Independent Gammas).

If $X_i \sim \text{Gamma}(\alpha_i, \beta)$ independently (same scale β), then $\sum X_i \sim \text{Gamma}(\sum \alpha_i, \beta)$.

Theorem 8.2 (Sum of Independent Chi-Squareds).

If $W_i \sim \chi^2(n_i)$ independently, then $\sum W_i \sim \chi^2(\sum n_i)$.

8.2 The Central Limit Theorem

Theorem 8.3 (Central Limit Theorem).

Let $X_1, X_2, \ldots$ be iid with mean μ and finite variance σ^2. Then

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} \mathcal{N}(0,1) \quad \text{as } n \to \infty.$$

Equivalently, for large n: $\bar{X} \mathrel{\dot{\sim}} \mathcal{N}(\mu, \sigma^2/n)$.

Example 8.1 (CLT Application (Lecture Sep 14)).

Problem. Roll a fair die $n = 100$ times. Find $\mathbb{P}(\bar{X} > 4)$.

Solution. $\mu = 3.5$, $\sigma^2 = 35/12 \approx 2.917$, $\text{SE} = \sigma/\sqrt{n} = \sqrt{2.917/100} \approx 0.1708$.

$$Z = \frac{4 - 3.5}{0.1708} = 2.928, \qquad \mathbb{P}(\bar{X} > 4) = \mathbb{P}(Z > 2.928) \approx \boxed{0.0017}.$$

Example 8.2 (CLT for Exponential (Lecture Sep 14)).

Problem. $X_i \overset{\text{iid}}{\sim} \text{Exp}(1)$, $n = 64$. Find $\mathbb{P}(0.9 \leq \bar{X} \leq 1.1)$.

Solution. $\mu = 1$, $\sigma = 1$, $\text{SE} = 1/\sqrt{64} = 0.125$.

$$Z_L = \frac{0.9 - 1}{0.125} = -0.8, \quad Z_U = \frac{1.1 - 1}{0.125} = 0.8.$$

$$\mathbb{P}(0.9 \leq \bar{X} \leq 1.1) = \Phi(0.8) - \Phi(-0.8) = 2(0.7881) - 1 = \boxed{0.5762}.$$

8.3 Sampling Distributions

Definition 8.1 (Student's t-Distribution).

If $Z \sim \mathcal{N}(0, 1)$ and $W \sim \chi^2(\nu)$ are independent, then

$$T = \frac{Z}{\sqrt{W/\nu}} \sim t_\nu.$$

In practice: $T = \dfrac{\bar{Y} - \mu}{S/\sqrt{n}} \sim t_{n-1}$.

Definition 8.2 (F-Distribution).

If $W_1 \sim \chi^2(\nu_1)$ and $W_2 \sim \chi^2(\nu_2)$ are independent, then

$$F = \frac{W_1/\nu_1}{W_2/\nu_2} \sim F_{\nu_1,\nu_2}.$$

Key Result 8.1 Reciprocal Property of the F-Distribution.

If $F \sim F_{\nu_1,\nu_2}$, then $1/F \sim F_{\nu_2,\nu_1}$. Also, $F_{1-\alpha,\nu_1,\nu_2} = 1/F_{\alpha,\nu_2,\nu_1}$.

Example 8.3 (F-Test for Equality of Variances (Session 2)).

Problem. Machine A: $n_1 = 16$, $S_1^2 = 4.2$. Machine B: $n_2 = 10$, $S_2^2 = 2.1$. Test $H_0 : \sigma_1^2 = \sigma_2^2$ at $\alpha = 0.05$.

Solution. Under H_0: $F = S_1^2/S_2^2 \sim F_{15,9}$.
Test statistic: $F = 4.2/2.1 = 2.0$.
Critical value: $F_{0.05,15,9} = 3.01$.
Since $2.0 < 3.01$: **Fail to reject** H_0. No significant difference in variances.

Solved Problem 8.1 (Identifying t and F Distributions (TT1, Fall 2024, Q2)).

Problem. $Y_1, \ldots, Y_5 \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$. Let $W = \sum_{i=1}^5 Y_i^2$ and $U = \sum_{i=1}^5 (Y_i - \bar{Y})^2$.
(a) Find the distribution of $\sqrt{5}\,\bar{Y}/\sqrt{W/5}$.
(b) Find the distribution of $5\bar{Y}^2/(U/4)$.

Solution.
(a) $\bar{Y} \sim \mathcal{N}(0, 1/5)$, so $\sqrt{5}\,\bar{Y} \sim \mathcal{N}(0,1)$. Also $W = \sum Y_i^2 \sim \chi^2(5)$.
But this is $\sqrt{5}\,\bar{Y}/\sqrt{W/5} = Z/\sqrt{W/5}$ where $Z \sim \mathcal{N}(0,1)$ and $W \sim \chi^2(5)$. However, Z and W are *not* independent (since $\bar{Y}$ is part of W). We rewrite:
$W = 5\bar{Y}^2 + U$ where $U = \sum (Y_i - \bar{Y})^2 \sim \chi^2(4)$ and $\bar{Y} \perp U$.
The ratio $\sqrt{5}\,\bar{Y}/\sqrt{U/4} \sim t_4$. But the denominator uses W, not U. The distribution of $\sqrt{5}\,\bar{Y}/\sqrt{W/5}$ does *not* simplify to a standard t-distribution.
However, $\boxed{\dfrac{\sqrt{5}\,\bar{Y}}{\sqrt{U/4}} \sim t_4}$ is the standard pivot.
(b) $5\bar{Y}^2 = (\sqrt{5}\,\bar{Y})^2 \sim \chi^2(1)$, and $U \sim \chi^2(4)$, independent.

$$\frac{5\bar{Y}^2/1}{U/4} = \frac{\chi^2(1)/1}{\chi^2(4)/4} \sim \boxed{F_{1,4}}.$$

Solved Problem 8.2 (F-distribution Probability (TT1 Review, Q7.36)).

Problem. $n_1 = 10$, $n_2 = 8$. If $\sigma_1^2 = 2\sigma_2^2$, find b such that $\mathbb{P}(S_1^2/S_2^2 \le b) = 0.95$.

Solution. $\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{9,7}$. So $\frac{S_1^2}{S_2^2} = \frac{\sigma_1^2}{\sigma_2^2} \cdot F_{9,7} = 2F_{9,7}$.

$$\mathbb{P}(S_1^2/S_2^2 \le b) = \mathbb{P}(2F_{9,7} \le b) = \mathbb{P}(F_{9,7} \le b/2) = 0.95.$$

From F-tables: $F_{0.95,9,7} = 3.68$, so $b/2 = 3.68$, giving $\boxed{b = 7.36}$.

Solved Problem 8.3 (CLT Probability (TT1 Review, Q7.50)).

Problem. $\sigma = 10$, $n = 100$. Find $\mathbb{P}(|\bar{Y} - \mu| \le 1)$.

Solution. $\bar{Y} \mathrel{\dot{\sim}} \mathcal{N}(\mu, 100/100) = \mathcal{N}(\mu, 1)$, so SE $= 1$.

$$\mathbb{P}(|\bar{Y} - \mu| \le 1) = \mathbb{P}(-1 \le Z \le 1) = 2\Phi(1) - 1 = 2(0.8413) - 1 = \boxed{0.6827}.$$

Example 8.4 (CLT for Apple Weights (Summer Session 3)).

Problem. Mean apple weight $\mu = 150$g, $\sigma = 80$g. Inspector samples $n = 36$. Find $\mathbb{P}(\bar{X} > 155)$.

Solution. By CLT: $\bar{X} \dot{\sim} \mathcal{N}(150, 80^2/36)$, SE $= 80/6 \approx 13.33$.

$$Z = \frac{155 - 150}{13.33} = 0.375, \quad \mathbb{P}(\bar{X} > 155) = 1 - \Phi(0.375) \approx \boxed{0.354}.$$

Example 8.5 (Distribution of $\bar{Y}$ and S^2 (Term Tests, Q1)).

Problem. $Y_1, \ldots, Y_5 \overset{\text{iid}}{\sim} \mathcal{N}(0, 1)$. Let $W = \sum Y_i^2$ and $U = \sum (Y_i - \bar{Y})^2$.
(a) Find the distribution of $\sqrt{5}\,\bar{Y}$.
(b) Find the distributions of W and U.

Solution.
(a) $\bar{Y} \sim \mathcal{N}(0, 1/5)$, so $\sqrt{5}\,\bar{Y} \sim \mathcal{N}(0, 1)$ and $(\sqrt{5}\,\bar{Y})^2 \sim \chi^2(1)$.
(b) $W = \sum Y_i^2$: each $Y_i^2 \sim \chi^2(1)$, so $W \sim \chi^2(5)$.
$U = \sum (Y_i - \bar{Y})^2$: by the sampling distribution result, $U/\sigma^2 = U \sim \chi^2(4)$.
Moreover, $W = (\sqrt{5}\,\bar{Y})^2 + U$, decomposing $\chi^2(5) = \chi^2(1) + \chi^2(4)$, with $\bar{Y}$ and U independent.

Key Result 8.2 Variance of the Sample Variance.

If $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$, then

$$\mathrm{Var}(S^2) = \frac{2\sigma^4}{n-1}.$$

Proof. Since $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$ and $\mathrm{Var}(\chi^2(k)) = 2k$:

$$\mathrm{Var}\left(\frac{(n-1)S^2}{\sigma^2}\right) = 2(n-1) \implies (n-1)^2 \frac{\mathrm{Var}(S^2)}{\sigma^4} = 2(n-1) \implies \mathrm{Var}(S^2) = \frac{2\sigma^4}{n-1}.$$

S Is Not Unbiased for σ Although S^2 is unbiased for σ^2, the sample standard deviation $S = \sqrt{S^2}$ is *not* unbiased for σ. By Jensen's inequality, $\mathbb{E}[S] = \mathbb{E}[\sqrt{S^2}] < \sqrt{\mathbb{E}[S^2]} = \sigma$. The bias is small for large n.

Chapter 9

Point Estimation

9.1 Introduction

Definition 9.1 (Point Estimator).

A **point estimator** $\hat{\theta} = \hat{\theta}(Y_1, \ldots, Y_n)$ is a function of the sample data used to estimate an unknown parameter θ.

9.2 Method of Moments

Definition 9.2 (Method of Moments (MoM)).

The Method of Moments equates sample moments to theoretical moments:

$$m_k = \frac{1}{n}\sum_{i=1}^{n} Y_i^k = \mathbb{E}[Y^k] = \mu_k'(\theta), \quad k = 1, 2, \ldots$$

Solve for θ in terms of $m_1, m_2, \ldots$

Example 9.1 (MoM for Uniform (Session 1)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta)$. Find the MoM estimator of θ.

Solution. $\mathbb{E}[Y] = \theta/2$. Set $\bar{Y} = \theta/2$ and solve: $\boxed{\hat{\theta}_{\text{MoM}} = 2\bar{Y}}$.

9.3 Maximum Likelihood Estimation

Definition 9.3 (Likelihood and Log-Likelihood).

The **likelihood function** for iid data $y_1, \ldots, y_n$ is

$$L(\theta) = \prod_{i=1}^{n} f(y_i; \theta).$$

The **log-likelihood** is $\ell(\theta) = \ln L(\theta) = \sum_{i=1}^{n} \ln f(y_i; \theta)$.

Definition 9.4 (Maximum Likelihood Estimator).

The **MLE** $\hat{\theta}_{\text{MLE}}$ is the value of θ that maximizes $L(\theta)$ (or equivalently $\ell(\theta)$).

Finding the MLE

1. Write the likelihood $L(\theta) = \prod f(y_i; \theta)$.
2. Take the log: $\ell(\theta) = \sum \ln f(y_i; \theta)$.
3. Differentiate: $\ell'(\theta) = 0$.
4. Solve for $\hat{\theta}$.
5. Verify it is a maximum (second derivative test or monotonicity).

If θ appears in the support, the MLE may occur at a boundary — use monotonicity instead.

9.3.1 MLE for Bernoulli

Example 9.2 (MLE for Bernoulli (MoM and MLE Notes)).

Problem. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$. Find $\hat{p}_{\text{MLE}}$.

Solution.

$$L(p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} = p^{\sum x_i}(1-p)^{n-\sum x_i}.$$

$$\ell(p) = \left(\sum x_i\right) \ln p + \left(n - \sum x_i\right) \ln(1-p).$$

$$\ell'(p) = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p} = 0.$$

Solve: $\sum x_i(1-p) = (n - \sum x_i)p$, giving $\boxed{\hat{p}_{\text{MLE}} = \bar{x} = \frac{1}{n}\sum x_i}$.

9.3.2 MLE for Poisson

Example 9.3 (MLE for Poisson (Q9.8)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$. Find $\hat{\lambda}_{\text{MLE}}$.

Solution.

$$L(\lambda) = \prod \frac{\lambda^{y_i} e^{-\lambda}}{y_i!} = \frac{\lambda^{\sum y_i} e^{-n\lambda}}{\prod y_i!}.$$

$$\ell(\lambda) = \left(\sum y_i\right) \ln \lambda - n\lambda - \sum \ln(y_i!).$$

$$\ell'(\lambda) = \frac{\sum y_i}{\lambda} - n = 0 \implies \boxed{\hat{\lambda}_{\text{MLE}} = \bar{y}}.$$

9.3.3 MLE for Exponential

Example 9.4 (MLE for Exponential (Q9.2)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$. Find $\hat{\theta}_{\text{MLE}}$.

Solution.

$$L(\theta) = \prod \frac{1}{\theta} e^{-y_i/\theta} = \theta^{-n} \exp\left(-\frac{\sum y_i}{\theta}\right).$$

$$\ell(\theta) = -n \ln \theta - \frac{\sum y_i}{\theta}.$$

$$\ell'(\theta) = -\frac{n}{\theta} + \frac{\sum y_i}{\theta^2} = 0 \implies \boxed{\hat{\theta}_{\text{MLE}} = \bar{y}}.$$

9.3.4 MLE for Gamma

Example 9.5 (MLE for Gamma with Known Shape (MLE Deep Notes, Q9.4a)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Gamma}(\alpha, \theta)$ with α known. Find $\hat{\theta}_{\text{MLE}}$.

Solution.

$$\ell(\theta) = -n\alpha \ln \theta + (\alpha - 1) \sum \ln y_i - \frac{\sum y_i}{\theta} - n \ln \Gamma(\alpha).$$

$$\ell'(\theta) = -\frac{n\alpha}{\theta} + \frac{\sum y_i}{\theta^2} = 0 \implies \boxed{\hat{\theta}_{\text{MLE}} = \frac{\sum y_i}{n\alpha} = \frac{\bar{y}}{\alpha}}.$$

Numerical: With $n = 3$, $\alpha = 2$, $\sum y_i = 378$: $\hat{\theta} = 378/6 = 63$ hours.

9.3.5 MLE for Uniform (Boundary Case)

Example 9.6 (MLE for Uniform — Boundary (Q9.3)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Uniform}(0, 2\theta + 1)$. Find $\hat{\theta}_{\text{MLE}}$.

Solution.

$$L(\theta) = (2\theta + 1)^{-n}, \quad 0 < y_i < 2\theta + 1 \text{ for all } i.$$

Since $\ell'(\theta) = -2n/(2\theta + 1) < 0$ for all θ, the likelihood is *decreasing* in θ.
To maximize, take the *smallest* θ satisfying the constraint $2\theta + 1 \geq Y_{(n)}$:

$$\boxed{\hat{\theta}_{\text{MLE}} = \frac{Y_{(n)} - 1}{2}}.$$

9.3.6 MLE for Rayleigh

Example 9.7 (MLE for Rayleigh (Term Tests Solved Again)).

Problem. Y_i has PDF $f(y;\beta) = \frac{2y}{\beta^2} e^{-y^2/\beta^2}$, $y > 0$. Find $\hat{\beta}_{\text{MLE}}$.

Solution.

$$\ell(\beta) = \sum \ln(2y_i) - 2n \ln \beta - \frac{\sum y_i^2}{\beta^2}.$$

$$\ell'(\beta) = -\frac{2n}{\beta} + \frac{2\sum y_i^2}{\beta^3} = 0 \implies n\beta^2 = \sum y_i^2 \implies \boxed{\hat{\beta}_{\text{MLE}} = \sqrt{\frac{\sum y_i^2}{n}}}.$$

9.3.7 MLE for Power Distribution (Q9.88, MLE Deep Notes)

Example 9.8 (MLE for $(\theta+1)y^\theta$ Distribution (MLE Deep, Q9.88)).

Problem. $Y_1, \ldots, Y_n$ iid with PDF $f(y;\theta) = (\theta+1)y^\theta$, $0 \le y \le 1$, $\theta > -1$.

Solution.

$$\ell(\theta) = n \ln(\theta+1) + \theta \sum \ln y_i.$$

$$\ell'(\theta) = \frac{n}{\theta+1} + \sum \ln y_i = 0 \implies \theta + 1 = \frac{-n}{\sum \ln y_i}.$$

$$\boxed{\hat{\theta}_{\text{MLE}} = \frac{-n}{\sum \ln Y_i} - 1}.$$

9.3.8 MLE for Multinomial Distribution (Q9.87, MLE Deep Notes)

Example 9.9 (MLE for Multinomial (MLE Deep, Q9.87)).

Problem. $n = 100$ observations classified: $n_1 = 30$ favor A (p_1), $n_2 = 38$ favor B (p_2), $n_3 = 32$ favor C (p_3). Find MLEs of p_1, p_2, p_3.

Solution. Multinomial likelihood:

$$L(p_1, p_2, p_3) = \frac{100!}{30!\,38!\,32!}\, p_1^{30}\, p_2^{38}\, p_3^{32}.$$

Using Lagrange multiplier for $p_1 + p_2 + p_3 = 1$:

$$\frac{\partial}{\partial p_i}\left[\ell - \lambda(p_1 + p_2 + p_3 - 1)\right] = 0 \implies \frac{n_i}{p_i} = \lambda \text{ for all } i.$$

So $p_i \propto n_i$. With the constraint: $\boxed{\hat{p}_1 = 0.30, \quad \hat{p}_2 = 0.38, \quad \hat{p}_3 = 0.32}$.
Part 2: MLE of $p_1 - p_2 = 0.30 - 0.38 = -0.08$ (by invariance).

9.3.9 MLE for Normal Distribution (Note Sep 25)

Example 9.10 (MLE for $\mathcal{N}(\mu, \sigma^2)$ (Note Sep 25)).

Problem. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. Find the MLEs of μ and σ^2.

Solution.

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\ln(2\pi\sigma^2) - \frac{\sum(x_i - \mu)^2}{2\sigma^2}.$$

Set partial derivatives to zero:

$$\frac{\partial \ell}{\partial \mu} = \frac{\sum(x_i - \mu)}{\sigma^2} = 0 \implies \boxed{\hat{\mu}_{\text{MLE}} = \bar{X}}.$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\sum(x_i - \mu)^2}{2\sigma^4} = 0 \implies \boxed{\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum(X_i - \bar{X})^2}.$$

Note: The MLE of σ^2 divides by n, not $n - 1$, so it is biased: $\mathbb{E}[\hat{\sigma}^2_{\text{MLE}}] = \frac{n-1}{n}\sigma^2$.

9.4 Properties of MLEs

9.4.1 MLE for Pareto Distribution

Example 9.11 (MLE for Pareto (Final Exam Fall 2025, Q4)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Pareto}(\theta)$ with PDF $f(y; \theta) = \theta / y^{\theta+1}$, $y > 1$, $\theta > 0$.
(a) Find $\hat{\theta}_{\text{MLE}}$.
(b) Find $\hat{\theta}_{\text{MoM}}$.

Solution.
(a) Log-likelihood:

$$\ell(\theta) = n \ln \theta - (\theta + 1) \sum \ln y_i.$$

$$\ell'(\theta) = \frac{n}{\theta} - \sum \ln y_i = 0 \implies \boxed{\hat{\theta}_{\text{MLE}} = \frac{n}{\sum \ln Y_i}}.$$

(b) For the Pareto distribution, $\mathbb{E}[Y] = \theta/(\theta - 1)$ when $\theta > 1$. Set $\bar{Y} = \theta/(\theta - 1)$:

$$\bar{Y}(\theta - 1) = \theta \implies \bar{Y}\theta - \bar{Y} = \theta \implies \theta(\bar{Y} - 1) = \bar{Y}.$$

$$\boxed{\hat{\theta}_{\text{MoM}} = \frac{\bar{Y}}{\bar{Y} - 1}}.$$

9.4.2 MLE for Beta Distribution

Example 9.12 (MLE for Beta$(\theta, 1)$ (Final Exam Fall 2024, Q5)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Beta}(\theta, 1)$ with PDF $f(y; \theta) = \theta y^{\theta - 1}$, $0 < y < 1$.
(a) Find $\hat{\theta}_{\text{MLE}}$.
(b) Show $\prod Y_i$ (equivalently $\sum \ln Y_i$) is sufficient.
(c) Find $\mathcal{I}(\theta)$ and the CRLB.

Solution.
(a) $\ell(\theta) = n \ln \theta + (\theta - 1) \sum \ln y_i$.

$$\ell'(\theta) = \frac{n}{\theta} + \sum \ln y_i = 0 \implies \boxed{\hat{\theta}_{\text{MLE}} = \frac{-n}{\sum \ln Y_i}}.$$

Note: $\ln Y_i < 0$ since $0 < Y_i < 1$, so $\hat{\theta} > 0$. ✓
(b) $L(\theta) = \theta^n \prod y_i^{\theta - 1} = \theta^n e^{(\theta - 1) \sum \ln y_i} = \underbrace{\theta^n e^{\theta \sum \ln y_i}}_{g(T, \theta)} \cdot \underbrace{e^{-\sum \ln y_i}}_{h(\boldsymbol{y})}$.
By Factorization, $T = \sum \ln Y_i$ is sufficient. ✓
(c) $\ln f = \ln \theta + (\theta - 1) \ln y$. $\frac{\partial^2}{\partial \theta^2} \ln f = -1/\theta^2$.

$$\boxed{\mathcal{I}(\theta) = \frac{1}{\theta^2}}, \quad \text{CRLB} = \frac{\theta^2}{n}.$$

9.4.3 MLE for Normal with Known Mean

Example 9.13 (MLE for $\mathcal{N}(0, \theta^2)$ (Final Exam Fall 2023, Q6)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(0, \theta^2)$ with unknown $\theta > 0$. Find $\hat{\theta}_{\text{MLE}}$.

Solution.

$$\ell(\theta) = -n \ln \theta - \frac{n}{2} \ln(2\pi) - \frac{\sum y_i^2}{2\theta^2}.$$

$$\ell'(\theta) = -\frac{n}{\theta} + \frac{\sum y_i^2}{\theta^3} = 0 \implies \theta^2 = \frac{\sum y_i^2}{n} \implies \boxed{\hat{\theta}_{\text{MLE}} = \sqrt{\frac{1}{n} \sum Y_i^2}}.$$

9.4.4 MLE for Shifted Exponential (Boundary Case)

Example 9.14 (MLE for Shifted Exponential (Final Fall 2023, Q5)).

Problem. $Y_1, \ldots, Y_n$ with PDF $f(y; \theta) = e^{-(y-\theta)}$, $y > \theta$. Find the MLE of θ.

Solution. $L(\theta) = \exp(-\sum(y_i - \theta)) = e^{n\theta} \cdot e^{-\sum y_i}$, subject to $\theta \leq y_i$ for all i, i.e., $\theta \leq Y_{(1)}$.
The log-likelihood $\ell(\theta) = n\theta - \sum y_i$ is *increasing* in θ.
Maximize by taking θ as large as possible: $\boxed{\hat{\theta}_{\text{MLE}} = Y_{(1)}}$.

Example 9.15 (MLE for Uniform$(0, 2\theta + 1)$ (MLE Deep, Q9.83)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Uniform}(0, 2\theta + 1)$. Find $\hat{\theta}_{\text{MLE}}$.

Solution. $L(\theta) = (2\theta + 1)^{-n}$ for $2\theta + 1 \geq y_{(n)}$, i.e., $\theta \geq (y_{(n)} - 1)/2$.
L is *decreasing* in θ. Maximize at smallest valid θ:

$$\boxed{\hat{\theta}_{\text{MLE}} = \frac{Y_{(n)} - 1}{2}}.$$

Example 9.16 (MLE for Truncated Power $f = 3y^2/\theta^3$ (MLE Deep, Q9.91)).

Problem. $Y_1, \ldots, Y_n$ with $f(y; \theta) = 3y^2/\theta^3$, $0 \leq y \leq \theta$. Find $\hat{\theta}_{\text{MLE}}$.

Solution. $L(\theta) = 3^n \prod y_i^2/\theta^{3n}$ for $\theta \geq y_{(n)}$.
$\ell(\theta) = n \ln 3 + 2 \sum \ln y_i - 3n \ln \theta$. $\ell'(\theta) = -3n/\theta < 0$ (decreasing).
Maximize at $\theta = y_{(n)}$: $\boxed{\hat{\theta}_{\text{MLE}} = Y_{(n)}}$.

Example 9.17 (MLE for Two-Sample Normal, Common Variance (MLE Deep, Q9.86)).

Problem. $X_1, \ldots, X_m \overset{\text{iid}}{\sim} \mathcal{N}(\mu_1, \sigma^2)$ and $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu_2, \sigma^2)$, independent. Find $\hat{\sigma}^2_{\text{MLE}}$.

Solution. Joint log-likelihood:

$$\ell = -\frac{m+n}{2}\ln(2\pi\sigma^2) - \frac{\sum(x_i-\mu_1)^2 + \sum(y_j-\mu_2)^2}{2\sigma^2}.$$

Setting $\partial\ell/\partial\mu_1 = 0$: $\hat{\mu}_1 = \bar{X}$. Similarly $\hat{\mu}_2 = \bar{Y}$.
Setting $\partial\ell/\partial\sigma^2 = 0$:

$$\boxed{\hat{\sigma}^2_{\text{MLE}} = \frac{\sum(X_i-\bar{X})^2 + \sum(Y_j-\bar{Y})^2}{m+n}}.$$

Note: this differs from the unbiased pooled estimator S_p^2 which divides by $m+n-2$.

Boundary MLEs When θ appears in the *support* of the distribution (e.g., $y > \theta$ or $0 < y < \theta$), the usual calculus approach (set derivative to zero) often fails. Instead, check whether the likelihood is monotone in θ and find the extremum at the boundary of the constraint region. Common cases:

- Uniform$(0, \theta)$: $\hat{\theta} = Y_{(n)}$ (maximum)
- Uniform$(\theta, \theta+1)$: boundary depends on both $Y_{(1)}$ and $Y_{(n)}$
- Shifted Exponential $f(y) = e^{-(y-\theta)}$, $y > \theta$: $\hat{\theta} = Y_{(1)}$ (minimum)

9.5 Properties of MLEs

Theorem 9.1 (Invariance Property of MLEs).

If $\hat{\theta}$ is the MLE of θ, then $\tau(\hat{\theta})$ is the MLE of $\tau(\theta)$ for any function τ.

Theorem 9.2 (Consistency of MLEs).

Under regularity conditions, $\hat{\theta}_{\text{MLE}} \xrightarrow{p} \theta$ as $n \to \infty$.

Example 9.18 (Proving Consistency via WLLN (Test 2, Fall 2023, Q3)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} f(y;\theta) = \theta y^{\theta-1}$, $0 < y < 1$. Show $\bar{Y}$ is a consistent estimator of $\theta/(\theta+1)$.

Solution. $Y \sim \text{Beta}(\theta, 1)$, so $\mathbb{E}[Y] = \theta/(\theta+1)$ and $\text{Var}(Y) = \theta/[(\theta+1)^2(\theta+2)]$ is finite. By the Weak Law of Large Numbers: $\bar{Y} \xrightarrow{p} \mathbb{E}[Y] = \theta/(\theta+1)$ as $n \to \infty$.

Therefore $\bar{Y}$ is consistent for $\theta/(\theta+1)$. $\square$

9.6 Bias, Variance, and MSE

Definition 9.5 (Bias).

The **bias** of an estimator $\hat{\theta}$ is

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta.$$

$\hat{\theta}$ is **unbiased** if $\text{Bias}(\hat{\theta}) = 0$, i.e., $\mathbb{E}[\hat{\theta}] = \theta$.

Definition 9.6 (Mean Squared Error).

$$\text{MSE}(\hat{\theta}) = \mathbb{E}[(\hat{\theta}-\theta)^2] = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2.$$

Theorem 9.3 (MSE Decomposition (Bias-Variance Tradeoff)).

$$\text{MSE}(\hat{\theta}) = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2.$$

Proof:

$$\begin{aligned}\text{MSE}(\hat{\theta}) &= \mathbb{E}[(\hat{\theta}-\theta)^2] = \mathbb{E}[(\hat{\theta}-\mathbb{E}[\hat{\theta}]+\mathbb{E}[\hat{\theta}]-\theta)^2] \\ &= \mathbb{E}[(\hat{\theta}-\mathbb{E}[\hat{\theta}])^2] + 2(\mathbb{E}[\hat{\theta}]-\theta)\underbrace{\mathbb{E}[\hat{\theta}-\mathbb{E}[\hat{\theta}]]}_{=0} + (\mathbb{E}[\hat{\theta}]-\theta)^2 \\ &= \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2.\end{aligned}$$

Definition 9.7 (Standard Error).

The **standard error** of $\hat{\theta}$ is $\text{SE}(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}$.

9.6.1 Solved Problems on Bias and MSE

Solved Problem 9.1 (Biased Estimator Correction (Lecture Sep 28, Ex. 8.5)).

Problem. Suppose $\mathbb{E}[\hat{\theta}] = a\theta + b$ for nonzero constants a, b.
(a) Find $\text{Bias}(\hat{\theta})$.
(b) Find a function of $\hat{\theta}$ that is unbiased for θ.

Solution.
(a) $\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta = a\theta + b - \theta = \boxed{(a-1)\theta + b}$.
(b) Let $\hat{\theta}^* = (\hat{\theta} - b)/a$. Then $\mathbb{E}[\hat{\theta}^*] = (a\theta + b - b)/a = \theta$. ✓

Solved Problem 9.2 (Optimal Convex Combination (Lecture Sep 28, Ex. 8.6)).

Problem. $\hat{\theta}_1, \hat{\theta}_2$ are independent unbiased estimators with $\mathrm{Var}(\hat{\theta}_1) = \sigma_1^2$, $\mathrm{Var}(\hat{\theta}_2) = \sigma_2^2$. Consider $\hat{\theta}_3 = a\hat{\theta}_1 + (1-a)\hat{\theta}_2$.
(a) Show $\hat{\theta}_3$ is unbiased.
(b) Find a that minimizes $\mathrm{Var}(\hat{\theta}_3)$.

Solution.
(a) $\mathbb{E}[\hat{\theta}_3] = a\theta + (1-a)\theta = \theta$. ✓
(b) By independence: $\mathrm{Var}(\hat{\theta}_3) = a^2\sigma_1^2 + (1-a)^2\sigma_2^2$.
Set the derivative to zero:

$$\frac{d}{da}\mathrm{Var}(\hat{\theta}_3) = 2a\sigma_1^2 - 2(1-a)\sigma_2^2 = 0.$$

$$a(\sigma_1^2 + \sigma_2^2) = \sigma_2^2 \implies \boxed{a = \frac{\sigma_2^2}{\sigma_1^2 + \sigma_2^2}}.$$

The weight is inversely proportional to the variance: more weight on the lower-variance estimator.

Solved Problem 9.3 (With Covariance (Lecture Sep 28, Ex. 8.7)).

Problem. Same setup but $\mathrm{Cov}(\hat{\theta}_1, \hat{\theta}_2) = c \neq 0$. Find optimal a.

Solution.

$$\mathrm{Var}(\hat{\theta}_3) = a^2\sigma_1^2 + (1-a)^2\sigma_2^2 + 2a(1-a)c.$$

$$\frac{d}{da} = 2a\sigma_1^2 - 2(1-a)\sigma_2^2 + 2c(1-2a) = 0.$$

Solving: $a(2\sigma_1^2 + 2\sigma_2^2 - 4c) = 2\sigma_2^2 - 2c$, so

$$\boxed{a = \frac{\sigma_2^2 - c}{\sigma_1^2 + \sigma_2^2 - 2c}}.$$

Solved Problem 9.4 ($N(0,\theta)$ Estimation (Summer 2025 Test 2)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(0,\theta)$ where $\theta > 0$ is the unknown variance. Let $\hat{\theta} = \frac{1}{n}\sum Y_i^2$.
(a) Show $\hat{\theta}$ is unbiased.
(b) Compute $\mathrm{MSE}(\hat{\theta})$.

Solution.
(a) $\mathbb{E}[\hat{\theta}] = \frac{1}{n}\sum \mathbb{E}[Y_i^2] = \frac{1}{n} \cdot n\theta = \theta$. ✓

(b) Since $\hat{\theta}$ is unbiased, $\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta})$.

$$\mathrm{Var}(\hat{\theta}) = \frac{1}{n^2} \sum \mathrm{Var}(Y_i^2) = \frac{\mathrm{Var}(Y^2)}{n}.$$

For $Y \sim \mathcal{N}(0, \theta)$: $\mathbb{E}[Y^2] = \theta$, $\mathbb{E}[Y^4] = 3\theta^2$ (fourth moment of normal), so

$$\mathrm{Var}(Y^2) = 3\theta^2 - \theta^2 = 2\theta^2.$$

$$\boxed{\mathrm{MSE}(\hat{\theta}) = \mathrm{Var}(\hat{\theta}) = \frac{2\theta^2}{n}}.$$

Chapter 10

Sufficient Statistics

10.1 Definition of Sufficiency

Definition 10.1 (Sufficient Statistic).

A statistic $T = T(Y_1, \ldots, Y_n)$ is **sufficient** for θ if the conditional distribution of the sample given T does not depend on θ.

10.2 Factorization Theorem

Theorem 10.1 (Fisher–Neyman Factorization Theorem).

A statistic $T(\boldsymbol{y})$ is sufficient for θ if and only if the likelihood can be written as

$$L(\theta) = g(T(\boldsymbol{y}), \theta) \cdot h(\boldsymbol{y}),$$

where g depends on the data *only through* T, and h does not depend on θ.

Applying the Factorization Theorem

1. Write the joint likelihood $L(\theta) = \prod f(y_i; \theta)$.
2. Identify terms that depend on θ and the data.
3. Factor into $g(T, \theta) \cdot h(\boldsymbol{y})$.
4. The statistic T appearing in g is sufficient.

Example 10.1 (Sufficiency for Bernoulli (Note Dec 11)).

Problem. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$. Show $T = \sum X_i$ is sufficient for p.

Solution.

$$L(p) = \prod p^{x_i}(1-p)^{1-x_i} = p^{\sum x_i}(1-p)^{n-\sum x_i} = \underbrace{p^T(1-p)^{n-T}}_{g(T,p)} \cdot \underbrace{1}_{h(\boldsymbol{x})}.$$

By the Factorization Theorem, $T = \sum X_i$ is sufficient for p. $\square$

Example 10.2 (Sufficiency for Normal Variance (Note Dec 11)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with μ known. Show $U = \sum (Y_i - \mu)^2$ is sufficient for σ^2.

Solution.

$$L(\sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum (y_i - \mu)^2}{2\sigma^2}\right) = \underbrace{(2\pi\sigma^2)^{-n/2} \exp\left(-\frac{U}{2\sigma^2}\right)}_{g(U,\sigma^2)} \cdot \underbrace{1}_{h(\boldsymbol{y})}.$$

By factorization, $U = \sum (Y_i - \mu)^2$ is sufficient for σ^2. $\square$

10.3 Exponential Families

Definition 10.2 (One-Parameter Exponential Family).

A family of distributions is a **one-parameter exponential family** if the PDF/PMF can be written as

$$f(y; \theta) = a(\theta)\, b(y)\, \exp[c(\theta)\, d(y)].$$

In this case, $T = \sum d(Y_i)$ is sufficient for θ.

Chapter 11

Completeness and Unbiased Estimation

11.1 Fisher Information

Definition 11.1 (Score Function).

The **score function** is the derivative of the log-likelihood for a single observation:

$$S(\theta) = \frac{\partial}{\partial\theta} \ln f(Y;\theta).$$

Key property: $\mathbb{E}[S(\theta)] = 0$.

Definition 11.2 (Fisher Information).

The **Fisher information** in a single observation is

$$\mathcal{I}(\theta) = \mathbb{E}\left[\left(\frac{\partial}{\partial\theta} \ln f(Y;\theta)\right)^2\right] = -\,\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \ln f(Y;\theta)\right].$$

For n iid observations: $\mathcal{I}_n(\theta) = n\,\mathcal{I}(\theta)$.

Remark 11.1. *The two formulas for Fisher information are equivalent under regularity conditions (interchange of differentiation and integration). Formula 2 (the expected negative second derivative) is usually easier to compute.*

Computing Fisher Information

1. Write $\ln f(y;\theta)$.
2. Compute $\frac{\partial^2}{\partial\theta^2} \ln f(y;\theta)$.
3. Take $-\,\mathbb{E}[\cdot]$ to get $\mathcal{I}(\theta)$.

11.2 Cramér–Rao Lower Bound

Theorem 11.1 (Cramér–Rao Inequality).

Let $\hat{\theta}$ be an unbiased estimator of θ. Under regularity conditions,

$$\mathrm{Var}(\hat{\theta}) \geq \frac{1}{n\,\mathcal{I}(\theta)}.$$

The right side is the **Cramér–Rao Lower Bound (CRLB)**.

Definition 11.3 (Efficient Estimator).

An unbiased estimator $\hat{\theta}$ is **efficient** if $\mathrm{Var}(\hat{\theta}) = \dfrac{1}{n\,\mathcal{I}(\theta)}$ (i.e., it achieves the CRLB).

Definition 11.4 (Efficiency).

The **efficiency** of an unbiased estimator $\hat{\theta}$ is

$$\mathrm{eff}(\hat{\theta}) = \frac{\mathrm{CRLB}}{\mathrm{Var}(\hat{\theta})} = \frac{1/(n\,\mathcal{I}(\theta))}{\mathrm{Var}(\hat{\theta})}.$$

Efficient $\iff$ $\mathrm{eff}(\hat{\theta}) = 1$.

Efficient $\Rightarrow$ MVUE, but not conversely An efficient estimator is automatically MVUE. However, an MVUE may exist even when no estimator achieves the CRLB. The CRLB is a *lower bound* — it need not be attained.

11.2.1 Fisher Information Examples

Example 11.1 (Fisher Information for Bernoulli (Note Nov 3)).

Problem. $Y \sim \mathrm{Bernoulli}(\theta)$. Find $\mathcal{I}(\theta)$.

Solution. $\ln f(y;\theta) = y\ln\theta + (1-y)\ln(1-\theta)$.

$$\frac{\partial^2}{\partial\theta^2}\ln f = -\frac{y}{\theta^2} - \frac{1-y}{(1-\theta)^2}.$$

$$\mathcal{I}(\theta) = -\,\mathbb{E}\Big[-\frac{Y}{\theta^2} - \frac{1-Y}{(1-\theta)^2}\Big] = \frac{\theta}{\theta^2} + \frac{1-\theta}{(1-\theta)^2} = \frac{1}{\theta} + \frac{1}{1-\theta} = \boxed{\frac{1}{\theta(1-\theta)}}.$$

CRLB: $\mathrm{Var}(\hat{\theta}) \geq \dfrac{\theta(1-\theta)}{n}$. Since $\mathrm{Var}(\bar{Y}) = \theta(1-\theta)/n$, the MLE $\bar{Y}$ is **efficient**.

Example 11.2 (Fisher Information for Poisson (Note Nov 3)).

Problem. $Y \sim \text{Poisson}(\theta)$. Find $\mathcal{I}(\theta)$ and check efficiency of $\bar{Y}$.

Solution. $\ln f(y;\theta) = y\ln\theta - \theta - \ln(y!)$.

$$\frac{\partial^2}{\partial\theta^2}\ln f = -\frac{y}{\theta^2}.$$

$$\mathcal{I}(\theta) = -\,\mathbb{E}\left[-\frac{Y}{\theta^2}\right] = \frac{\mathbb{E}[Y]}{\theta^2} = \frac{\theta}{\theta^2} = \boxed{\frac{1}{\theta}}.$$

CRLB: θ/n. Since $\text{Var}(\bar{Y}) = \theta/n = \text{CRLB}$, $\bar{Y}$ is **efficient** and is the MVUE.

Example 11.3 (Fisher Information for Exponential (Fisher Notes)).

Problem. $Y \sim \text{Exp}(\theta)$. Find $\mathcal{I}(\theta)$ and check efficiency of $\bar{Y}$.

Solution. $\ln f = -\ln\theta - y/\theta$.

$$\frac{\partial}{\partial\theta}\ln f = -\frac{1}{\theta} + \frac{y}{\theta^2}, \qquad \frac{\partial^2}{\partial\theta^2}\ln f = \frac{1}{\theta^2} - \frac{2y}{\theta^3}.$$

$$\mathcal{I}(\theta) = -\,\mathbb{E}\left[\frac{1}{\theta^2} - \frac{2Y}{\theta^3}\right] = -\frac{1}{\theta^2} + \frac{2\theta}{\theta^3} = \boxed{\frac{1}{\theta^2}}.$$

CRLB: θ^2/n. Since $\text{Var}(\bar{Y}) = \theta^2/n = \text{CRLB}$, $\bar{Y}$ is **efficient**.

Example 11.4 (Fisher Information for Gamma (MLE Deep Notes)).

Problem. $Y \sim \text{Gamma}(\alpha,\theta)$ with α known. Find $\mathcal{I}(\theta)$ and efficiency of $\hat{\theta} = \bar{Y}/\alpha$.

Solution. $\ln f = -\alpha\ln\theta + (\alpha-1)\ln y - y/\theta - \ln\Gamma(\alpha)$.

$$\frac{\partial^2}{\partial\theta^2}\ln f = \frac{\alpha}{\theta^2} - \frac{2y}{\theta^3}.$$

$$\mathcal{I}(\theta) = -\frac{\alpha}{\theta^2} + \frac{2\alpha\theta}{\theta^3} = \boxed{\frac{\alpha}{\theta^2}}.$$

CRLB: $\theta^2/(n\alpha)$. Estimator: $\text{Var}(\bar{Y}/\alpha) = \text{Var}(\bar{Y})/\alpha^2 = \alpha\theta^2/(n\alpha^2) = \theta^2/(n\alpha) = \text{CRLB}$.
So $\bar{Y}/\alpha$ is **efficient** — eff $= 1$.

Example 11.5 (Fisher Information for Normal Variance (Fisher Notes)).

Problem. $Y \sim \mathcal{N}(\mu,\sigma^2)$ with μ known, σ^2 unknown. Find $\mathcal{I}(\sigma^2)$.

Solution. Let $r = \sigma^2$. $\ln f = -\frac{1}{2}\ln(2\pi r) - \frac{(y-\mu)^2}{2r}$.

$$\frac{\partial}{\partial r}\ln f = -\frac{1}{2r} + \frac{(y-\mu)^2}{2r^2}, \qquad \frac{\partial^2}{\partial r^2}\ln f = \frac{1}{2r^2} - \frac{(y-\mu)^2}{r^3}.$$

$$\mathcal{I}(\sigma^2) = -\mathbb{E}\left[\frac{1}{2r^2} - \frac{(Y-\mu)^2}{r^3}\right] = -\frac{1}{2\sigma^4} + \frac{\sigma^2}{\sigma^6} = \boxed{\frac{1}{2\sigma^4}}.$$

CRLB for σ^2: $2\sigma^4/n$. Since $\mathrm{Var}\left(\frac{1}{n}\sum(Y_i-\mu)^2\right) = 2\sigma^4/n$, the estimator $\frac{1}{n}\sum(Y_i-\mu)^2$ achieves the CRLB and is the MVUE of σ^2 when μ is known.

11.3 Relative Efficiency

Definition 11.5 (Relative Efficiency).

Given two unbiased estimators $\hat{\theta}_1, \hat{\theta}_2$, the **relative efficiency** of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is

$$\mathrm{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\mathrm{Var}(\hat{\theta}_2)}{\mathrm{Var}(\hat{\theta}_1)}.$$

If > 1: $\hat{\theta}_1$ is more efficient. If < 1: $\hat{\theta}_2$ is more efficient.

Solved Problem 11.1 (Comparing Uniform Estimators (Final Exam Fall 2025, Q2)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathrm{Uniform}(0, \theta)$. Two unbiased estimators: $\hat{\theta}_1 = 2\bar{Y}$, $\hat{\theta}_2 = \dfrac{n+1}{n}Y_{(n)}$.
(a) Show both are unbiased. (b) Compute the efficiency of $\hat{\theta}_1$ relative to $\hat{\theta}_2$.

Solution.
(a) $\mathbb{E}[\hat{\theta}_1] = 2\,\mathbb{E}[\bar{Y}] = 2 \cdot \theta/2 = \theta$. ✓
For $Y_{(n)}$: $\mathbb{E}[Y_{(n)}] = n\theta/(n+1)$, so $\mathbb{E}[\hat{\theta}_2] = \frac{n+1}{n} \cdot \frac{n\theta}{n+1} = \theta$. ✓
(b) Variances:

$$\mathrm{Var}(\hat{\theta}_1) = 4\,\mathrm{Var}(\bar{Y}) = 4 \cdot \frac{\theta^2}{12n} = \frac{\theta^2}{3n}.$$

For $Y_{(n)}$: $\mathrm{Var}(Y_{(n)}) = \frac{n\theta^2}{(n+1)^2(n+2)}$, so

$$\mathrm{Var}(\hat{\theta}_2) = \left(\frac{n+1}{n}\right)^2 \mathrm{Var}(Y_{(n)}) = \frac{(n+1)^2}{n^2} \cdot \frac{n\theta^2}{(n+1)^2(n+2)} = \frac{\theta^2}{n(n+2)}.$$

Efficiency:

$$\mathrm{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\mathrm{Var}(\hat{\theta}_2)}{\mathrm{Var}(\hat{\theta}_1)} = \frac{\theta^2/[n(n+2)]}{\theta^2/(3n)} = \frac{3}{n+2}.$$

For $n \geq 2$: $3/(n+2) < 1$, so $\hat{\theta}_2$ is **more efficient**. The order-statistic estimator dominates the sample-mean estimator. $\hat{\theta}_2$ is preferred.

11.4 Minimum Variance Unbiased Estimators (MVUE)

Theorem 11.2 (Rao–Blackwell Theorem).

If $\hat{\theta}$ is any unbiased estimator and T is sufficient for θ, then $\hat{\theta}^* = \mathbb{E}[\hat{\theta} \mid T]$ is also unbiased and $\mathrm{Var}(\hat{\theta}^*) \leq \mathrm{Var}(\hat{\theta})$.

Theorem 11.3 (Lehmann–Scheffé Theorem).

If T is a complete sufficient statistic and $h(T)$ is unbiased for θ, then $h(T)$ is the unique MVUE of θ.

Solved Problem 11.2 (MVUE for Rayleigh (Note Dec 11, Q9.58)).

Problem. Y_i has Rayleigh distribution with $Y_i^2 \sim \mathrm{Exp}(\theta)$. $U = \sum Y_i^2$ is sufficient. Find the MVUE of θ.

Solution. Since $Y_i^2 \sim \mathrm{Exp}(\theta)$, $U = \sum Y_i^2 \sim \mathrm{Gamma}(n, \theta)$, so $\mathbb{E}[U] = n\theta$.
Therefore $\mathbb{E}[U/n] = \theta$, and U/n is an unbiased function of the sufficient statistic U.

$$\boxed{\hat{\theta}_{\mathrm{MVUE}} = \frac{1}{n} \sum Y_i^2}.$$

Solved Problem 11.3 (MVUE for Two Normal Samples (Note Dec 11, Q9.57)).

Problem. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu_X, \sigma^2)$ and $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu_Y, \sigma^2)$ independent, common variance σ^2 unknown. Is $\hat{\sigma}^2 = \frac{\sum(X_i - \bar{X})^2 + \sum(Y_i - \bar{Y})^2}{2n-2}$ an MVUE?

Solution.
Unbiasedness:

$$\mathbb{E}[\hat{\sigma}^2] = \frac{(n-1)\sigma^2 + (n-1)\sigma^2}{2n-2} = \frac{2(n-1)\sigma^2}{2(n-1)} = \sigma^2. \checkmark$$

Since it is an unbiased function of the complete sufficient statistic $\sum(X_i - \bar{X})^2 + \sum(Y_i - \bar{Y})^2$, by Lehmann–Scheffé it is the **MVUE** of σ^2.

Solved Problem 11.4 (MVUE of σ^2 with Known Mean (Note Dec 11, Q9.56)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ where μ is *known*. Find the MVUE of σ^2.

Solution. Since μ is known, $U = \sum(Y_i - \mu)^2$ is sufficient for σ^2.
Distribution: $(Y_i - \mu)/\sigma \sim \mathcal{N}(0, 1)$, so $(Y_i - \mu)^2/\sigma^2 \sim \chi^2(1)$, hence $U/\sigma^2 \sim \chi^2(n)$.
Therefore $\mathbb{E}[U] = n\sigma^2$, so $\mathbb{E}[U/n] = \sigma^2$.

$$\boxed{\hat{\sigma}^2_{\text{MVUE}} = \frac{1}{n}\sum(Y_i - \mu)^2}.$$

Note: With μ *known*, we divide by n (not $n-1$), since we use the $\chi^2(n)$ distribution instead of $\chi^2(n-1)$.

Solved Problem 11.5 (MVUE of $p(1-p)$ for Bernoulli (Final Fall 2025, Q5c)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$. Use Rao–Blackwell to find the MVUE of $\tau(p) = p(1-p)$.

Solution. The complete sufficient statistic is $T = \sum Y_i \sim \text{Bin}(n, p)$.
Start with the unbiased estimator $\hat{\theta} = Y_1(1 - Y_2)$ (since $\mathbb{E}[Y_1(1 - Y_2)] = p(1-p)$ by independence).
By Rao–Blackwell, compute $\hat{\theta}^* = \mathbb{E}[Y_1(1 - Y_2) \mid T]$.
Key insight: $\mathbb{E}[Y_1Y_2 \mid T = t] = \mathbb{P}(Y_1 = 1, Y_2 = 1 \mid T = t) = \frac{\binom{n-2}{t-2}}{\binom{n}{t}} = \frac{t(t-1)}{n(n-1)}$.
Also $\mathbb{E}[Y_1 \mid T] = T/n$. Therefore:

$$\hat{\theta}^* = \mathbb{E}[Y_1 \mid T] - \mathbb{E}[Y_1Y_2 \mid T] = \frac{T}{n} - \frac{T(T-1)}{n(n-1)} = \frac{T(n-1) - T(T-1)}{n(n-1)} = \frac{T(n-T)}{n(n-1)}.$$

$$\boxed{\text{MVUE of } p(1-p) = \frac{T(n-T)}{n(n-1)}, \quad T = \sum Y_i}.$$

Solved Problem 11.6 (Completeness and MVUE for Gamma-type (Final Fall 2024, Q3)).

Problem. $Y_1, \ldots, Y_n$ from $f(y; \theta) = \frac{1}{\theta^2}\, y\, e^{-y/\theta}$, $y > 0$ (i.e., Gamma$(2, \theta)$).
(a) Show $\sum Y_i$ is complete and sufficient.
(b) Show $\hat{\theta} = \frac{1}{2n}\sum Y_i$ is the unique MVUE of θ.

Solution.
(a) Write $f(y; \theta) = \frac{y}{\theta^2}\, e^{-y/\theta}$. This is an exponential family with $d(y) = y$, $c(\theta) = -1/\theta$.
By the exponential family theorem, $T = \sum Y_i$ is complete and sufficient for θ.
(b) Since $Y_i \sim \text{Gamma}(2, \theta)$, by the sum-of-gammas theorem: $T = \sum Y_i \sim \text{Gamma}(2n, \theta)$, so $\mathbb{E}[T] = 2n\theta$.

Therefore $\mathbb{E}[T/(2n)] = \theta$, so $\hat{\theta} = T/(2n) = \frac{1}{2n}\sum Y_i$ is unbiased.
Since $\hat{\theta}$ is an unbiased function of the complete sufficient statistic T, by **Lehmann–Scheffé**, it is the unique MVUE. □

Solved Problem 11.7 (Beta MVUE via Rao–Blackwell (Final Fall 2025, Q3)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Beta}(\theta, 1)$, $f(y;\theta) = \theta y^{\theta-1}$, $0 < y < 1$. The complete sufficient statistic is $T = \sum \ln Y_i$ (or $\prod Y_i$). Derive the MVUE of θ.

Solution. Since $-\ln Y_i \sim \text{Exp}(1/\theta)$, we have $-T = -\sum \ln Y_i \sim \text{Gamma}(n, 1/\theta)$.
So $\mathbb{E}[-T] = n/\theta$, giving $\mathbb{E}[(-T)/n] = 1/\theta$, hence $\mathbb{E}[n/(-T)] \neq \theta$ (the reciprocal is biased).
Instead, for $\text{Gamma}(n, 1/\theta)$, we know that $\mathbb{E}[1/(-T)] = \theta/(n-1)$ for $n > 1$.
Therefore $\mathbb{E}[(n-1)/(-T)] = \theta$, so

$$\boxed{\hat{\theta}_{\text{MVUE}} = \frac{n-1}{-\sum \ln Y_i} = \frac{-(n-1)}{\sum \ln Y_i}}.$$

This is the unique MVUE by Lehmann–Scheffé, and it equals $\frac{n-1}{n} \cdot \hat{\theta}_{\text{MLE}}$.

Solved Problem 11.8 (MVUE for Power Distribution $f = 3y^2/\theta^3$ (Note Dec 16, Q9.63)).

Problem. $Y_1, \ldots, Y_n$ with $f(y;\theta) = 3y^2/\theta^3$, $0 \le y \le \theta$. Find the MVUE of θ.

Solution. CDF: $F(y) = y^3/\theta^3$. By Factorization, $Y_{(n)}$ is sufficient (θ in the support).
PDF of $Y_{(n)}$: $f_{Y_{(n)}}(y) = 3ny^{3n-1}/\theta^{3n}$, $0 \le y \le \theta$.

$$\mathbb{E}[Y_{(n)}] = \frac{3n}{\theta^{3n}} \int_0^\theta y^{3n}\,\mathrm{d}y = \frac{3n}{\theta^{3n}} \cdot \frac{\theta^{3n+1}}{3n+1} = \frac{3n}{3n+1}\,\theta.$$

Bias-correct: $\mathbb{E}\left[\frac{3n+1}{3n} Y_{(n)}\right] = \theta$. By Lehmann–Scheffé:

$$\boxed{\hat{\theta}_{\text{MVUE}} = \frac{3n+1}{3n}\,Y_{(n)}}.$$

Solved Problem 11.9 (MVUE for σ^2 with Known μ (Note Dec 11, Q9.57)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with μ *known*. Find the MVUE of σ^2.

Solution. $U = \sum (Y_i - \mu)^2$ is sufficient. Since $(Y_i - \mu)/\sigma \overset{\text{iid}}{\sim} \mathcal{N}(0,1)$, $U/\sigma^2 \sim \chi^2(n)$, so $\mathbb{E}[U] = n\sigma^2$.

Therefore $\mathbb{E}[U/n] = \sigma^2$:

$$\boxed{\hat{\sigma}^2_{\text{MVUE}} = \frac{1}{n}\sum(Y_i - \mu)^2}.$$

Note: this uses n in the denominator (not $n-1$) because μ is *known* — no degrees of freedom lost for estimating the mean.

Solved Problem 11.10 (Gamma$(2, \beta)$ CRLB Verification (Final Fall 2025 B, Q3)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Gamma}(2, \beta)$. Verify the UMVUE $\hat{\beta} = \bar{Y}/2$ attains the CRLB.

Solution. Single-observation Fisher info:

$$\ln f = -2\ln\beta + \ln y - y/\beta - \ln\Gamma(2).$$

$\frac{\partial^2}{\partial\beta^2}\ln f = \frac{2}{\beta^2} - \frac{2y}{\beta^3}$, so $\mathcal{I}(\beta) = -\frac{2}{\beta^2} + \frac{2\,\mathbb{E}[Y]}{\beta^3} = -\frac{2}{\beta^2} + \frac{4\beta}{\beta^3} = \frac{2}{\beta^2}$.
CRLB $= \beta^2/(2n)$.
$\text{Var}(\bar{Y}/2) = \text{Var}(\bar{Y})/4 = \frac{\alpha\beta^2}{4n} = \frac{2\beta^2}{4n} = \frac{\beta^2}{2n} = \text{CRLB}$. ✓
So $\bar{Y}/2$ is efficient and is the MVUE.

Solved Problem 11.11 (MVUE of Poisson Cost Function (Note Dec 11, Q9.59)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$. Breakdowns cost $C = 3Y^2$. Find the MVUE of $\mathbb{E}[C] = 3\,\mathbb{E}[Y^2] = 3(\lambda + \lambda^2)$.

Solution. The sufficient statistic is $U = \sum Y_i \sim \text{Poisson}(n\lambda)$, so $\mathbb{E}[U] = n\lambda$ and $\mathbb{E}[U^2] = n\lambda + n^2\lambda^2$.
We need an unbiased function $h(U)$ such that $\mathbb{E}[h(U)] = 3\lambda + 3\lambda^2$.
Write $h(U) = aU + bU^2$ and match moments:

$$\mathbb{E}[aU + bU^2] = an\lambda + b(n\lambda + n^2\lambda^2) = (an + bn)\lambda + bn^2\lambda^2.$$

Set equal to $3\lambda + 3\lambda^2$:

$$bn^2 = 3 \implies b = 3/n^2; \qquad (an + bn) = 3 \implies a = \frac{3 - 3/n}{n} = \frac{3(n-1)}{n^2}.$$

$$\boxed{\text{MVUE of } \mathbb{E}[C] = \frac{3(n-1)}{n^2}U + \frac{3}{n^2}U^2 = \frac{3}{n^2}U(n - 1 + U)}, \quad U = \sum Y_i.$$

Solved Problem 11.12 (MVUE for Uniform$(0, \theta + 1)$ (Note Dec 11, Q9.61)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta + 1)$. Show that $Y_{(n)} = \max(Y_1, \ldots, Y_n)$ is sufficient and find the MVUE of θ.

Solution. $L(\theta) = (\theta+1)^{-n} \cdot \mathbf{1}(Y_{(n)} \le \theta+1)$. By factorization, $Y_{(n)}$ is sufficient.
The PDF of $Y_{(n)}$ on Uniform$(0, \theta+1)$: $f_{Y_{(n)}}(y) = ny^{n-1}/(\theta+1)^n$, $0 \le y \le \theta+1$.

$$\mathbb{E}[Y_{(n)}] = \frac{n}{(\theta+1)^n} \int_0^{\theta+1} y^n \, \mathrm{d}y = \frac{n(\theta+1)}{n+1}.$$

So $\mathbb{E}\left[\frac{n+1}{n} Y_{(n)}\right] = \theta+1$, hence $\mathbb{E}\left[\frac{n+1}{n} Y_{(n)} - 1\right] = \theta$.

$$\boxed{\hat{\theta}_{\text{MVUE}} = \frac{n+1}{n} Y_{(n)} - 1}.$$

Solved Problem 11.13 (Bias and MSE of Shifted Exponential Minimum (TUT 4, Q1–Q2)).

Problem. $X_1, \ldots, X_n$ iid with PDF $f(x;\mu) = e^{-(x-\mu)}$, $x \ge \mu$. Let $\hat{\mu} = X_{(1)} = \min(X_1, \ldots, X_n)$.
(a) Find Bias$(\hat{\mu})$.
(b) Find MSE$(\hat{\mu})$.

Solution.
(a) $X_{(1)}$ has PDF $f_{X_{(1)}}(x) = ne^{-n(x-\mu)}$, $x \ge \mu$. So $X_{(1)} - \mu \sim \text{Exp}(1/n)$:

$$\mathbb{E}[X_{(1)}] = \mu + \frac{1}{n}, \qquad \text{Bias}(\hat{\mu}) = \mathbb{E}[\hat{\mu}] - \mu = \boxed{\frac{1}{n}}.$$

The bias-corrected estimator is $\hat{\mu}^* = X_{(1)} - 1/n$.
(b) $\text{Var}(X_{(1)}) = \text{Var}(\text{Exp}(1/n)) = 1/n^2$.

$$\text{MSE}(\hat{\mu}) = \text{Var}(\hat{\mu}) + [\text{Bias}(\hat{\mu})]^2 = \frac{1}{n^2} + \frac{1}{n^2} = \boxed{\frac{2}{n^2}}.$$

Solved Problem 11.14 (Comparing Three Exponential Estimators (Relative Efficiency Notes, Ex. 9.2)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$. Compare the estimators:

$$\hat{A}_1 = \frac{Y_1 + Y_2}{2}, \quad \hat{A}_2 = \frac{Y_1}{4} + \frac{\sum_{i=2}^{n-1} Y_i}{2(n-2)} + \frac{Y_n}{4}, \quad \hat{A}_3 = \bar{Y}.$$

Solution. All three are unbiased (verify by linearity of expectation; each is a convex combination with weights summing to 1). Using $\text{Var}(Y_i) = \theta^2$:
$\text{Var}(\hat{A}_1) = \frac{1}{4}(2\theta^2) = \theta^2/2$.
$\text{Var}(\hat{A}_2) = \frac{1}{16}\theta^2 + \frac{(n-2)\theta^2}{4(n-2)^2} + \frac{1}{16}\theta^2 = \frac{\theta^2}{8} + \frac{\theta^2}{4(n-2)}$.
$\text{Var}(\hat{A}_3) = \theta^2/n$.

Relative efficiency of $\hat{A}_1$ to $\hat{A}_3$:

$$\text{eff}(\hat{A}_1, \hat{A}_3) = \frac{\text{Var}(\hat{A}_3)}{\text{Var}(\hat{A}_1)} = \frac{\theta^2/n}{\theta^2/2} = \frac{2}{n}.$$

For $n > 2$: $\hat{A}_3 = \bar{Y}$ is more efficient. The sample mean beats all the ad hoc estimators.

Chapter 12

Hypothesis Testing

12.1 Framework

Definition 12.1 (Hypotheses).

- **Null hypothesis** H_0: the default claim (e.g., $\theta = \theta_0$).
- **Alternative hypothesis** H_A: the competing claim (e.g., $\theta \neq \theta_0$, $\theta > \theta_0$, or $\theta = \theta_1$).
- **Simple hypothesis**: specifies θ exactly. **Composite**: specifies a range.

Definition 12.2 (Type I and Type II Errors).

- **Type I error** (α): Reject H_0 when H_0 is true. $\alpha = \mathbb{P}(\text{reject } H_0 \mid H_0)$.
- **Type II error** (β): Fail to reject H_0 when H_A is true. $\beta = \mathbb{P}(\text{fail to reject } H_0 \mid H_A)$.
- **Power**: $1 - \beta = \mathbb{P}(\text{reject } H_0 \mid H_A)$.

12.2 Neyman–Pearson Lemma

Theorem 12.1 (Neyman–Pearson Lemma).

For testing $H_0 : \theta = \theta_0$ vs $H_A : \theta = \theta_1$ (both simple), the **most powerful test** of size α rejects H_0 when

$$\frac{L(\theta_0)}{L(\theta_1)} \leq k,$$

where k is chosen so that $\mathbb{P}(\text{reject} \mid H_0) = \alpha$.

Example 12.1 (Neyman–Pearson for Normal Mean (Test 2, Fall 2023, Q7)).

Problem. $Y \sim f(y;\theta) = 1 - \theta^2(y - 0.5)$ on $[0, 1]$. Test $H_0 : \theta = 0$ vs $H_A : \theta = 1/2$, based on a single observation Y.

Solution. Under H_0: $f(y; 0) = 1$ (i.e., $Y \sim \text{Uniform}(0, 1)$).
Under H_A: $f(y; 1/2) = 1 - \frac{1}{4}(y - 0.5)$.
Likelihood ratio:

$$\frac{L(0)}{L(1/2)} = \frac{1}{1 - \frac{1}{4}(y - 0.5)} \leq k.$$

This ratio is decreasing in y when $y < 0.5$ and increasing when $y > 0.5$. Analysis shows the

rejection region is RR : $\{Y \le c\}$ where c is chosen so $\mathbb{P}(Y \le c \mid H_0) = \alpha$.
Under H_0, $Y \sim \text{Uniform}(0,1)$, so $c = \alpha$.
Power: $\mathbb{P}(Y \le \alpha \mid H_A) = \int_0^\alpha [1 - \frac{1}{4}(y - 0.5)]\,\mathrm{d}y$.

Solved Problem 12.1 (Neyman–Pearson for Normal Mean (Final Fall 2025, Q6)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 4)$ ($\sigma^2 = 4$ known). Test $H_0 : \mu = 10$ vs $H_A : \mu = 12$, size α.
(a) Find the rejection region.
(b) For $n = 16$, $\alpha = 0.05$: find the critical value and compute the power.

Solution.
(a) Likelihood ratio:

$$\frac{L(\mu_0 = 10)}{L(\mu_1 = 12)} = \exp\left(-\frac{\sum(y_i - 10)^2}{8} + \frac{\sum(y_i - 12)^2}{8}\right) = \exp\left(\frac{n(12^2 - 10^2) - 2\sum y_i(12 - 10)}{8}\right).$$

Simplifying: the ratio is $\le k$ iff $\bar{Y} \ge c$ for some constant c. So the rejection region is RR : $\{\bar{Y} \ge c\}$.
Under H_0: $\bar{Y} \sim \mathcal{N}(10, 4/n)$. To get size α:

$$c = 10 + z_\alpha \cdot \frac{2}{\sqrt{n}}.$$

(b) With $n = 16$, $z_{0.05} = 1.645$:

$$c = 10 + 1.645 \cdot \frac{2}{4} = 10 + 0.8225 = 10.8225.$$

Power: Under H_A ($\mu = 12$): $\bar{Y} \sim \mathcal{N}(12, 4/16) = \mathcal{N}(12, 0.25)$.

$$\text{Power} = \mathbb{P}(\bar{Y} \ge 10.8225 \mid \mu = 12) = \mathbb{P}\left(Z \ge \frac{10.8225 - 12}{0.5}\right) = \mathbb{P}(Z \ge -2.355) = \Phi(2.355) \approx \boxed{0.991}.$$

Solved Problem 12.2 (Neyman–Pearson for Normal Mean, $\sigma = 1$ (Final Fall 2024, Q6)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 1)$. Test $H_0 : \mu = 0$ vs $H_A : \mu = 5$.
(a) Find the rejection region in terms of $\bar{Y}$.
(b) Compute the power for general n and α.

Solution.
(a) By the same NP analysis, the rejection region is RR : $\{\bar{Y} \ge c\}$ where

$$c = 0 + z_\alpha/\sqrt{n} = z_\alpha/\sqrt{n}.$$

(b) Under H_A ($\mu = 5$): $\bar{Y} \sim \mathcal{N}(5, 1/n)$.

$$\text{Power} = \mathbb{P}\left(\bar{Y} \geq \frac{z_\alpha}{\sqrt{n}} \,\middle|\, \mu = 5\right) = \mathbb{P}\left(Z \geq \frac{z_\alpha/\sqrt{n} - 5}{1/\sqrt{n}}\right) = \mathbb{P}(Z \geq z_\alpha - 5\sqrt{n}).$$

$$\boxed{\text{Power} = 1 - \Phi(z_\alpha - 5\sqrt{n})}.$$

For large n, $z_\alpha - 5\sqrt{n} \to -\infty$, so power $\to 1$.

General Recipe for Neyman–Pearson Problems

1. Write the likelihood ratio $L(\theta_0)/L(\theta_1)$.
2. Simplify and determine: reject when $\bar{Y} \geq c$ (if $\theta_1 > \theta_0$) or $\bar{Y} \leq c$ (if $\theta_1 < \theta_0$).
3. Find c using the null distribution and significance level α.
4. Compute power by evaluating $\mathbb{P}(\text{reject} \mid H_A)$ using the alternative distribution.

12.3 Power Function

Definition 12.3 (Power Function).

The **power function** of a test is $\pi(\theta) = \mathbb{P}(\text{reject } H_0 \mid \theta)$.
Under H_0: $\pi(\theta_0) = \alpha$ (Type I error rate).
Under H_A: $\pi(\theta_1) = 1 - \beta$ (power).
A good test has α small and power large.

12.4 Likelihood Ratio Tests

Definition 12.4 (Likelihood Ratio Test Statistic).

$$\Lambda = \frac{\sup_{\theta \in \Theta_0} L(\theta)}{\sup_{\theta \in \Theta} L(\theta)},$$

where Θ_0 is the parameter space under H_0. Reject H_0 when $\Lambda \leq c$.

Key Result 12.1 Wilks' Theorem (Large-Sample LRT).

Under H_0 and regularity conditions, $-2 \ln \Lambda \xrightarrow{d} \chi^2(r)$ as $n \to \infty$, where $r = \dim(\Theta) - \dim(\Theta_0)$ is the number of restricted parameters.

12.5 Common Hypothesis Tests

Key Result 12.2 Standard Test Statistics.

Z-test (σ known): $Z = \dfrac{\bar{Y} - \mu_0}{\sigma/\sqrt{n}} \sim \mathcal{N}(0,1)$ under H_0.

One-sample t-test (σ unknown): $T = \dfrac{\bar{Y} - \mu_0}{S/\sqrt{n}} \sim t_{n-1}$ under H_0.

Two-sample t-test (equal variances): $T = \dfrac{\bar{Y}_1 - \bar{Y}_2}{S_p\sqrt{1/n_1 + 1/n_2}} \sim t_{n_1+n_2-2}$.

χ^2-test for variance: $\chi^2 = \dfrac{(n-1)S^2}{\sigma_0^2} \sim \chi^2(n-1)$ under $H_0 : \sigma^2 = \sigma_0^2$.

F-test for ratio of variances: $F = S_1^2/S_2^2 \sim F_{n_1-1,n_2-1}$ under $H_0 : \sigma_1^2 = \sigma_2^2$.

Solved Problem 12.3 (NP Test — Best Critical Region (Final Fall 2022, Q3)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 100)$ ($\sigma^2 = 100$ known). Test $H_0 : \mu = 75$ vs $H_1 : \mu = 78$. Show $C = \{(x_1, \ldots, x_n) : \bar{x} \geq c\}$ is a best critical region. Find n and c such that $\alpha = 0.05$ and power $= 0.90$.

Solution. By NP lemma, the best critical region rejects when the likelihood ratio is small:

$$\frac{L(75)}{L(78)} = \exp\left(-\frac{\sum(x_i - 75)^2 - \sum(x_i - 78)^2}{200}\right).$$

Simplifying the exponent: this is $\leq k$ iff $\bar{x} \geq c$. ✓

Finding n and c:

Under H_0: $\bar{X} \sim \mathcal{N}(75, 100/n)$.

$$\alpha = 0.05: \quad c = 75 + 1.645 \cdot \frac{10}{\sqrt{n}}.$$

Under H_1: $\bar{X} \sim \mathcal{N}(78, 100/n)$.

$$\text{Power} = 0.90: \quad \mathbb{P}(\bar{X} \geq c \mid \mu = 78) = 0.90 \implies c = 78 - 1.282 \cdot \frac{10}{\sqrt{n}}.$$

Setting equal:

$$75 + \frac{16.45}{\sqrt{n}} = 78 - \frac{12.82}{\sqrt{n}} \implies \frac{29.27}{\sqrt{n}} = 3 \implies n = \left(\frac{29.27}{3}\right)^2 \approx \boxed{95.2 \implies n = 96}.$$

Then $c = 75 + 16.45/\sqrt{96} \approx \boxed{76.68}$.

Chapter 13

Interval Estimation and Confidence Intervals

13.1 Pivotal Quantities

Definition 13.1 (Pivotal Quantity).

A **pivotal quantity** $Q = Q(Y_1, \ldots, Y_n; \theta)$ is a function of the data and θ whose distribution does not depend on θ.

Constructing CIs via Pivots

1. Identify a pivot Q with known distribution.
2. Find quantiles: $\mathbb{P}(q_L \leq Q \leq q_U) = 1 - \alpha$.
3. Invert to isolate θ: rearrange to get $\theta_L \leq \theta \leq \theta_U$.

13.2 Confidence Intervals for Normal Parameters

Key Result 13.1 Standard Confidence Intervals.

For μ (σ known): $\bar{Y} \pm z_{\alpha/2} \dfrac{\sigma}{\sqrt{n}}$.

For μ (σ unknown): $\bar{Y} \pm t_{\alpha/2,n-1} \dfrac{S}{\sqrt{n}}$.

For σ^2: $\left[\dfrac{(n-1)S^2}{\chi^2_{\alpha/2,n-1}}, \dfrac{(n-1)S^2}{\chi^2_{1-\alpha/2,n-1}}\right]$.

For $\mu_1 - \mu_2$ (equal variances): $(\bar{Y}_1 - \bar{Y}_2) \pm t_{\alpha/2,n_1+n_2-2} \cdot S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$,

where $S_p^2 = \dfrac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}$.

For proportion p: $\hat{p} \pm z_{\alpha/2} \sqrt{\dfrac{\hat{p}(1-\hat{p})}{n}}$.

13.2.1 Worked Examples from Notes

Example 13.1 (CI for Mean, σ Unknown (Session 2)).

Problem. Sample of $n = 9$ students, $\bar{X} = 82$, $S = 6$. Construct a 95% CI for μ.

Solution. $df = 8$, $t_{0.025,8} = 2.306$.

$$82 \pm 2.306 \cdot \frac{6}{\sqrt{9}} = 82 \pm 2.306(2) = 82 \pm 4.612.$$

$$\boxed{95\% \text{ CI} : (77.39, 86.61)}.$$

Example 13.2 (CI for Difference in Means (Lecture 6, SAT Scores)).

Problem. $n_1 = 15$ (Engineering), $n_2 = 15$ (Language). $\bar{X}_1 = 446$, $\bar{X}_2 = 534$, $s_1 = 42$, $s_2 = 45$. Construct 95% CI.

Solution. $S_p^2 = \frac{14(42)^2+14(45)^2}{28} = \frac{24696+28350}{28} = 1894.5$. $S_p = 43.53$.
$df = 28$, $t_{0.025,28} = 2.048$.

$$(446 - 534) \pm 2.048 \cdot 43.53\sqrt{\frac{1}{15} + \frac{1}{15}} = -88 \pm 2.048(43.53)(0.3651) = -88 \pm 32.55.$$

$$\boxed{95\% \text{ CI} : (-120.55, -55.45)}.$$

Engineering students score 55–121 points lower on verbal SAT.

Example 13.3 (CI for Variance (Lecture 6)).

Problem. $n = 3$, data: 4.1, 5.2, 10.2. Construct 90% CI for σ^2.

Solution. $\bar{Y} = 6.5$, $S^2 = [(4.1 - 6.5)^2 + (5.2 - 6.5)^2 + (10.2 - 6.5)^2]/2 = 10.57$.
$df = 2$, $\chi^2_{0.05,2} = 5.991$, $\chi^2_{0.95,2} = 0.103$.

$$\left[\frac{2(10.57)}{5.991}, \frac{2(10.57)}{0.103}\right] = \boxed{(3.53, 205.24)}.$$

Example 13.4 (CI for Proportion (Session 2, Romance Movies)).

Problem. $n = 800$, $\hat{p} = 0.45$. Construct 98% CI for p.

Solution. $z_{0.01} = 2.326$.

$$0.45 \pm 2.326\sqrt{\frac{0.45(0.55)}{800}} = 0.45 \pm 2.326(0.0176) = 0.45 \pm 0.041.$$

$$\boxed{98\% \text{ CI} : (0.409, 0.491)}.$$

Since 0.50 is not in the interval, there is no evidence that a majority believe movies are getting better.

Solved Problem 13.1 (Pivot CI for Beta-type (Test 2, Fall 2023, Q4)).

Problem. Let $U = Y/\theta$ with $f_U(u) = 2u$, $0 \le u \le 1$. Find a two-sided $(1-\alpha)\%$ CI for θ.

Solution. $F_U(u) = u^2$.
Find quantiles: $F_U(u_L) = \alpha/2 \Rightarrow u_L = \sqrt{\alpha/2}$; $F_U(u_U) = 1 - \alpha/2 \Rightarrow u_U = \sqrt{1-\alpha/2}$.

$$\mathbb{P}\left(\sqrt{\alpha/2} \le \frac{Y}{\theta} \le \sqrt{1-\alpha/2}\right) = 1 - \alpha.$$

Invert:

$$\boxed{\left[\frac{Y}{\sqrt{1-\alpha/2}}, \frac{Y}{\sqrt{\alpha/2}}\right]}.$$

13.3 Sample Size Determination

Key Result 13.2 Sample Size Formulas.

For means: $n = \left(\frac{z_{\alpha/2} \cdot \sigma}{B}\right)^2$ where B is the desired bound on error.

For proportions: $n = \left(\frac{z_{\alpha/2}}{B}\right)^2 p(1-p)$. Use $p = 0.5$ for conservative estimate.

Example 13.5 (Sample Size (Lecture Sep 14)).

Problem. Want $\mathbb{P}(|\bar{X} - \mu| \le 0.2) \ge 0.95$ with $\sigma = 1$.

Solution. $z_{0.025} = 1.96$. Set $B = 0.2$:

$$n = \left(\frac{1.96 \times 1}{0.2}\right)^2 = 96.04 \implies \boxed{n = 97}.$$

Solved Problem 13.2 (Gunpowder Muzzle Velocity CI (Ch 8 Muzz Session)).

Problem. $n = 8$. Data: 3005, 2925, 2935, 2965, 2995, 3005, 2937, 2905. Construct a 95% CI for the true mean muzzle velocity.

Solution. $\bar{y} = 23672/8 = 2959$. $S = 39.1$. $df = 7$, $t_{0.025,7} = 2.365$.

$$2959 \pm 2.365 \cdot \frac{39.1}{\sqrt{8}} = 2959 \pm 2.365(13.83) = 2959 \pm 32.7.$$

$$\boxed{95\% \text{ CI} : (2926.3, 2991.7) \text{ ft/s}}.$$

Solved Problem 13.3 (Verbal vs. Math SAT CI (Note Jul 21, Q8.83)).

Problem. Verbal: $n_1 = 10$, $\bar{X}_1 = 505$, $S_1 = 57$. Math: $n_2 = 10$, $\bar{X}_2 = 495$, $S_2 = 69$. Construct a 95% CI for $\mu_1 - \mu_2$ (equal variances).

Solution. $S_p^2 = \frac{9(57)^2+9(69)^2}{18} = \frac{9(3249)+9(4761)}{18} = \frac{72090}{18} = 4005$. $S_p = 63.29$.
Wait — let me recompute. $S_p^2 = (3249 + 4761)/2 = 4005$. $S_p = 63.29$.
$df = 18$, $t_{0.025,18} = 2.101$.

$$(505 - 495) \pm 2.101 \cdot 63.29\sqrt{\frac{2}{10}} = 10 \pm 2.101(63.29)(0.4472) = 10 \pm 59.5.$$

$$\boxed{95\% \text{ CI} : (-49.5, 69.5)}.$$

Since 0 is in the interval, no significant difference in verbal vs. math scores.

13.4 Confidence Intervals for Two Populations

Key Result 13.3 Large-Sample CIs.

For $n_1, n_2 \geq 30$:

Difference of means: $(\bar{Y}_1 - \bar{Y}_2) \pm z_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$.

Difference of proportions: $(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$.

Key Result 13.4 CI for Ratio of Variances.

$$\left[\frac{S_1^2/S_2^2}{F_{\alpha/2,n_1-1,n_2-1}}, \frac{S_1^2/S_2^2}{F_{1-\alpha/2,n_1-1,n_2-1}}\right].$$

13.4.1 Additional Worked Examples

Example 13.6 (Two-Sample CI for Common Variance (Final Fall 2024, Q2)).

Problem. Two independent normal samples with common variance σ^2. Use the pivotal quantity $T = \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{\sigma^2} \sim \chi^2(n_1+n_2-2)$ to construct a CI for σ^2.

Solution. Let $\nu = n_1 + n_2 - 2$ and $T = [(n_1-1)S_1^2 + (n_2-1)S_2^2]/\sigma^2$.

$$\mathbb{P}\Big(\chi^2_{1-\alpha/2,\nu} \le T \le \chi^2_{\alpha/2,\nu}\Big) = 1-\alpha.$$

Invert:

$$\boxed{\left[\frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{\chi^2_{\alpha/2,\nu}}, \frac{(n_1-1)S_1^2+(n_2-1)S_2^2}{\chi^2_{1-\alpha/2,\nu}}\right]}.$$

Solved Problem 13.4 (CI from Minimum Order Statistic (Final Fall 2025, Q1c)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$. Given $Q = 2nY_{(1)}/\theta \sim \chi^2(2)$, derive a $(1-\alpha)100\%$ CI for θ.

Solution. Since $Q \sim \chi^2(2)$:

$$\mathbb{P}\left(\chi^2_{1-\alpha/2,2} \le \frac{2nY_{(1)}}{\theta} \le \chi^2_{\alpha/2,2}\right) = 1-\alpha.$$

Invert:

$$\boxed{\left[\frac{2nY_{(1)}}{\chi^2_{\alpha/2,2}}, \frac{2nY_{(1)}}{\chi^2_{1-\alpha/2,2}}\right]}.$$

Solved Problem 13.5 (Pivot CI for Beta-type (Test 2, Fall 2023, Q4)).

Problem. Single observation Y from $f(y;\theta) = 2y/\theta^2$, $0 \le y \le \theta$. Let $U = Y/\theta$. The PDF of U is $f_U(u) = 2u$, $0 \le u \le 1$, with $F_U(u) = u^2$. Construct a $(1-\alpha)100\%$ CI for θ.

Solution. Find quantiles: $F_U(u_L) = \alpha/2 \implies u_L = \sqrt{\alpha/2}$; $F_U(u_U) = 1-\alpha/2 \implies u_U = \sqrt{1-\alpha/2}$.

$$\mathbb{P}\left(\sqrt{\alpha/2} \le \frac{Y}{\theta} \le \sqrt{1-\alpha/2}\right) = 1-\alpha.$$

Invert: $Y/\sqrt{1-\alpha/2} \le \theta \le Y/\sqrt{\alpha/2}$.

$$\boxed{\left[\frac{Y}{\sqrt{1-\alpha/2}}, \frac{Y}{\sqrt{\alpha/2}}\right]}.$$

Solved Problem 13.6 (CI for Question 8.80 (Note Jul 21)).

Problem. $n = 21$, $\bar{X} = 2.6$, $S = 7.4$. Construct a 95% CI for μ.

Solution. $df = 20$, $t_{0.025,20} = 2.086$.

$$2.6 \pm 2.086 \cdot \frac{7.4}{\sqrt{21}} = 2.6 \pm 2.086(1.614) = 2.6 \pm 3.37.$$

$$\boxed{95\% \text{ CI} : (-0.77, 5.97)}.$$

Since 0 is in the interval, we cannot conclude that the true mean differs from 0.

Solved Problem 13.7 (CI Length for Normal Mean (Test 2, Fall 2022, Q1)).

Problem. $X_1, \ldots, X_{10} \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$ with σ known. Find the length of a 95% CI for μ.

Solution. The pivot is $\sqrt{n}(\bar{X} - \mu)/\sigma \sim \mathcal{N}(0, 1)$. CI: $\bar{X} \pm z_{0.025}\sigma/\sqrt{n}$.

$$\text{Length} = 2z_{0.025} \cdot \frac{\sigma}{\sqrt{n}} = 2(1.96) \cdot \frac{\sigma}{\sqrt{10}} = \boxed{1.239\sigma}.$$

Solved Problem 13.8 (Paired Sample CI (Test 2, Fall 2022, Q2)).

Problem. Height measurements of 5 children at ages 3 and 5:

Child	3 yrs (x_i)	5 yrs (y_i)	$d_i = y_i - x_i$
1	95.6	114.8	19.2
2	75.4	84.0	8.6
3	76.6	99.7	23.1
4	99.4	121.0	21.6
5	65.7	80.7	15.0

$\bar{d} = 17.5$, $S_d^2 = 34.1$. Construct a 95% CI for $\Delta = \mu_2 - \mu_1$.

Solution. Paired t-interval with $df = n - 1 = 4$, $t_{0.025,4} = 2.776$.

$$\bar{d} \pm t_{0.025,4} \cdot \frac{S_d}{\sqrt{n}} = 17.5 \pm 2.776 \cdot \frac{\sqrt{34.1}}{\sqrt{5}} = 17.5 \pm 2.776(2.612) = 17.5 \pm 7.25.$$

$$\boxed{95\% \text{ CI} : (10.25, 24.75)}.$$

Since 0 is not in the interval, the children grow significantly between ages 3 and 5.

Solved Problem 13.9 (Two-Sample CI — Reading Comprehension (Test 2, Fall 2021, Q3)).

Problem. Two methods for teaching reading: $n_1 = 12$, $\bar{y}_1 = 64$, $S_1^2 = 52$ and $n_2 = 13$, $\bar{y}_2 = 69$, $S_2^2 = 71$. Construct a 95% CI for $\mu_1 - \mu_2$ (equal variances assumed).

Solution. $S_p^2 = \frac{11(52)+12(71)}{23} = \frac{572+852}{23} = 61.913$, $S_p = 7.868$.
$df = 23$, $t_{0.025,23} = 2.069$.

$$(64 - 69) \pm 2.069 \cdot 7.868\sqrt{\frac{1}{12} + \frac{1}{13}} = -5 \pm 2.069(7.868)(0.4006) = -5 \pm 6.517.$$

$$\boxed{95\% \text{ CI} : (-11.52, 1.52)}.$$

Since 0 is in the interval, we cannot conclude the methods differ significantly.

Chapter 14

Order Statistics

14.1 Definition and Notation

Definition 14.1 (Order Statistics).

Given a random sample $Y_1, \ldots, Y_n$, the **order statistics** are the sample values arranged in increasing order: $Y_{(1)} \leq Y_{(2)} \leq \cdots \leq Y_{(n)}$.
$Y_{(1)} = \min(Y_1, \ldots, Y_n)$ is the **sample minimum**.
$Y_{(n)} = \max(Y_1, \ldots, Y_n)$ is the **sample maximum**.

14.2 Distribution of the Maximum

Theorem 14.1 (Distribution of $Y_{(n)}$).

If $Y_1, \ldots, Y_n$ are iid with CDF $F(y)$ and PDF $f(y)$, then:

$$F_{Y_{(n)}}(y) = [F(y)]^n, \qquad f_{Y_{(n)}}(y) = n[F(y)]^{n-1} f(y).$$

Example 14.1 (Maximum of Uniform (Session 4)).

Problem. $Y_1, Y_2, Y_3 \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta)$. Find the PDF of $Y_{(3)}$.

Solution. $F(y) = y/\theta$ for $0 \leq y \leq \theta$.

$$f_{Y_{(3)}}(y) = 3\left(\frac{y}{\theta}\right)^2 \cdot \frac{1}{\theta} = \boxed{\frac{3y^2}{\theta^3}}, \quad 0 \leq y \leq \theta.$$

14.3 Distribution of the Minimum

Theorem 14.2 (Distribution of $Y_{(1)}$).

$$F_{Y_{(1)}}(y) = 1 - [1 - F(y)]^n, \qquad f_{Y_{(1)}}(y) = n[1 - F(y)]^{n-1} f(y).$$

Example 14.2 (Minimum of Exponential (Practice Order Stats, Q4)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$ (mean λ). Find the distribution of $Y_{(1)}$.

Solution. $F(y) = 1 - e^{-y/\lambda}$ for $y > 0$.

$$f_{Y_{(1)}}(y) = n\left[e^{-y/\lambda}\right]^{n-1} \cdot \frac{1}{\lambda} e^{-y/\lambda} = \frac{n}{\lambda} e^{-ny/\lambda}, \quad y > 0.$$

This is $\text{Exp}(\lambda/n)$, so $\boxed{Y_{(1)} \sim \text{Exp}(\lambda/n)}$ with $\mathbb{E}[Y_{(1)}] = \lambda/n$.

Example 14.3 (Minimum of Shifted Exponential (Note Dec 16)).

Problem. Y_i have shifted exponential: $f(y;\theta) = e^{-(y-\theta)}$, $y \geq \theta$. Find $\mathbb{E}[Y_{(1)}]$ and an unbiased estimator based on $Y_{(1)}$.

Solution. $F(y) = 1 - e^{-(y-\theta)}$ for $y \geq \theta$.

$$f_{Y_{(1)}}(y) = ne^{-n(y-\theta)}, \quad y \geq \theta.$$

This is a shifted exponential with rate n and shift θ, so $Y_{(1)} - \theta \sim \text{Exp}(1/n)$.

$$\mathbb{E}[Y_{(1)}] = \theta + \frac{1}{n}.$$

Therefore $\boxed{\hat{\theta} = Y_{(1)} - 1/n}$ is unbiased for θ. Since $Y_{(1)}$ is sufficient, this is the MVUE.

Solved Problem 14.1 (MVUE from Minimum of Uniform (Note Dec 16)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Uniform}(\theta, \theta+1)$. Find the MVUE of θ based on $Y_{(1)}$.

Solution. $F(y) = y - \theta$ for $\theta \leq y \leq \theta + 1$.

$$f_{Y_{(1)}}(y) = n(\theta + 1 - y)^{n-1}, \quad \theta \leq y \leq \theta + 1.$$

Substituting $u = y - \theta$:

$$\mathbb{E}[Y_{(1)}] = \int_{\theta}^{\theta+1} y \cdot n(\theta + 1 - y)^{n-1}\, \mathrm{d}y = \theta + \frac{1}{n+1}.$$

Therefore $\boxed{\hat{\theta}_{\text{MVUE}} = Y_{(1)} - \frac{1}{n+1}}$.

Solved Problem 14.2 (Comparing Uniform Estimators (Tutorial 4, Q3)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Uniform}(\theta, \theta+1)$. Show that both $\hat{\theta}_1 = \bar{Y} - 1/2$ and $\hat{\theta}_2 = Y_{(n)} - n/(n+1)$ are unbiased.

Solution.
$\hat{\theta}_1$: $\mathbb{E}[\bar{Y}] = \theta + 1/2$, so $\mathbb{E}[\bar{Y} - 1/2] = \theta$. ✓
$\hat{\theta}_2$: The maximum has PDF $f_{Y_{(n)}}(y) = n(y-\theta)^{n-1}$ for $\theta \le y \le \theta + 1$.

$$\mathbb{E}[Y_{(n)}] = \int_\theta^{\theta+1} y \cdot n(y-\theta)^{n-1}\,\mathrm{d}y = \theta + \frac{n}{n+1}.$$

So $\mathbb{E}[Y_{(n)} - n/(n+1)] = \theta$. ✓
Both are unbiased for θ.

14.4 Distribution of the k-th Order Statistic

Theorem 14.3 (PDF of $Y_{(k)}$).

$$f_{Y_{(k)}}(y) = \frac{n!}{(k-1)!(n-k)!}[F(y)]^{k-1}[1-F(y)]^{n-k} f(y).$$

14.5 Expected Values and Variances of Order Statistics

Key Result 14.1 Order Statistics of Uniform$(0, \theta)$.

If $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta)$, then for $Y_{(n)} = \max$:

$$\mathbb{E}[Y_{(n)}] = \frac{n}{n+1}\theta, \quad \text{Var}(Y_{(n)}) = \frac{n\theta^2}{(n+1)^2(n+2)}.$$

For $Y_{(1)} = \min$:

$$\mathbb{E}[Y_{(1)}] = \frac{\theta}{n+1}, \quad \text{Var}(Y_{(1)}) = \frac{n\theta^2}{(n+1)^2(n+2)}.$$

Solved Problem 14.3 (Complete Order Stat Analysis for Uniform (Q6.74)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta)$. Find the PDF of $Y_{(n)}$, $\mathbb{E}[Y_{(n)}]$, and $\text{Var}(Y_{(n)})$.

Solution. $F(y) = y/\theta$ for $0 \le y \le \theta$.

$$f_{Y_{(n)}}(y) = n\left(\frac{y}{\theta}\right)^{n-1} \cdot \frac{1}{\theta} = \frac{ny^{n-1}}{\theta^n}, \quad 0 \le y \le \theta.$$

$\mathbb{E}[Y_{(n)}] = \frac{n}{\theta^n}\int_0^\theta y^n\,\mathrm{d}y = \frac{n}{\theta^n}\cdot\frac{\theta^{n+1}}{n+1} = \boxed{\dfrac{n\theta}{n+1}}$.

$\mathbb{E}[Y_{(n)}^2] = \frac{n}{\theta^n}\int_0^\theta y^{n+1}\,\mathrm{d}y = \frac{n\theta^2}{n+2}$.

$\mathrm{Var}(Y_{(n)}) = \frac{n\theta^2}{n+2} - \frac{n^2\theta^2}{(n+1)^2} = \boxed{\dfrac{n\theta^2}{(n+1)^2(n+2)}}$.

Solved Problem 14.4 (Minimum of Exponential — Full Analysis (Q6.84)).

Problem. $Y_1,\ldots,Y_n \overset{\text{iid}}{\sim} \mathrm{Exp}(\beta)$. Find the distribution of $Y_{(1)}$ and compute $\mathbb{P}(Y_{(1)} \le 3.6)$ for $n = 5$, $\beta = 2$.

Solution. $F(y) = 1 - e^{-y/\beta}$, $y > 0$.

$$F_{Y_{(1)}}(y) = 1 - [1 - F(y)]^n = 1 - e^{-ny/\beta}.$$

So $Y_{(1)} \sim \mathrm{Exp}(\beta/n)$ with $\mathbb{E}[Y_{(1)}] = \beta/n$.
For $n = 5$, $\beta = 2$: $Y_{(1)} \sim \mathrm{Exp}(2/5)$.

$$\mathbb{P}(Y_{(1)} \le 3.6) = 1 - e^{-5\cdot 3.6/2} = 1 - e^{-9} \approx \boxed{0.99988}.$$

Solved Problem 14.5 (Maximum of Exponential — PDF (Practice Order Stats, Q2)).

Problem. $Y_1,\ldots,Y_n \overset{\text{iid}}{\sim} \mathrm{Exp}(\omega)$ (rate ω). Find the PDF of $Y_{(n)} = \max$.

Solution. $F(y) = 1 - e^{-\omega y}$ for $y > 0$.

$$f_{Y_{(n)}}(y) = n[1 - e^{-\omega y}]^{n-1}\cdot\omega e^{-\omega y}, \quad y > 0.$$

$\boxed{f_{Y_{(n)}}(y) = n\omega e^{-\omega y}(1 - e^{-\omega y})^{n-1}}$.
Note: unlike the minimum, the maximum of exponentials does *not* follow a simple named distribution.

Solved Problem 14.6 ($\mathbb{E}[1/Y]$ for Cubic Distribution Max (Practice Order Stats, Q5)).

Problem. X_1, X_2 iid with $f_X(x) = 3x^2$, $0 < x < 1$. Let $Y = \max(X_1, X_2)$. Find $\mathbb{E}[1/Y]$.

Solution. CDF: $F_X(x) = x^3$. PDF of max:

$$f_Y(y) = 2[F_X(y)]^1 f_X(y) = 2(y^3)(3y^2) = 6y^5, \quad 0 < y < 1.$$

$$\mathbb{E}\left[\frac{1}{Y}\right] = \int_0^1 \frac{1}{y} \cdot 6y^5 \, \mathrm{d}y = \int_0^1 6y^4 \, \mathrm{d}y = 6 \cdot \frac{y^5}{5}\bigg|_0^1 = \boxed{\frac{6}{5}}.$$

Solved Problem 14.7 ($\mathrm{Var}(2Y_{(1)} + 1)$ for Exponential Min (Practice Order Stats, Q4c)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathrm{Exp}(\lambda)$ (mean λ). $Y_{(1)} \sim \mathrm{Exp}(\lambda/n)$. Find $\mathrm{Var}(2Y_{(1)} + 1)$.

Solution. Since $Y_{(1)} \sim \mathrm{Exp}(\lambda/n)$, $\mathrm{Var}(Y_{(1)}) = (\lambda/n)^2 = \lambda^2/n^2$.

$$\mathrm{Var}(2Y_{(1)} + 1) = 4\,\mathrm{Var}(Y_{(1)}) = \boxed{\frac{4\lambda^2}{n^2}}.$$

Solved Problem 14.8 (Maximum of Uniform$(\theta, \theta+1)$ — PDF (Term Test 1, 2024, Q3)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathrm{Uniform}(\theta, \theta + 1)$. Find the PDF of $Y_{(n)} = \max$.

Solution. $F(y) = y - \theta$ for $\theta \le y \le \theta + 1$, $f(y) = 1$.

$$f_{Y_{(n)}}(y) = n[F(y)]^{n-1} f(y) = \boxed{n(y-\theta)^{n-1}}, \quad \theta \le y \le \theta + 1.$$

Solved Problem 14.9 (Hypothesis Test via Order Statistic (Test 2, Fall 2022, Q3)).

Problem. $X_1, \ldots, X_4 \overset{\text{iid}}{\sim} \mathrm{Uniform}(0, \theta)$. Test $H_0 : \theta = 1$ vs $H_1 : \theta > 1$. Reject H_0 if $Y_{(4)} \ge c$.
(a) Find c such that $\alpha = 0.05$.
(b) Find the power function.

Solution.
(a) $Y_{(4)}$ has PDF $f(y) = 4y^3/\theta^4$ on $[0, \theta]$. Under H_0 ($\theta = 1$): $f(y) = 4y^3$ on $[0, 1]$.

$$\alpha = \mathbb{P}(Y_{(4)} \ge c \mid \theta = 1) = \int_c^1 4y^3 \, \mathrm{d}y = 1 - c^4 = 0.05.$$

$$c^4 = 0.95 \implies \boxed{c = 0.95^{1/4} \approx 0.9873}.$$

(b) For $\theta > 1$:

$$\pi(\theta) = \mathbb{P}(Y_{(4)} \ge c \mid \theta) = \int_c^\theta \frac{4y^3}{\theta^4} \, \mathrm{d}y = \frac{\theta^4 - c^4}{\theta^4} = \boxed{1 - \left(\frac{c}{\theta}\right)^4}.$$

Note: $\pi(1) = 0.05$ (size) and $\pi(\theta) \to 1$ as $\theta \to \infty$.

14.6 Estimation Using Order Statistics

Order-Statistic Based Estimators For Uniform$(0, \theta)$: the MLE is $\hat{\theta}_{\text{MLE}} = Y_{(n)}$, but this is biased ($\mathbb{E}[Y_{(n)}] = n\theta/(n+1) \neq \theta$). The bias-corrected version $\hat{\theta} = \frac{n+1}{n} Y_{(n)}$ is unbiased and has variance $\theta^2/[n(n+2)]$, which is smaller than $\text{Var}(2\bar{Y}) = \theta^2/(3n)$.

Solved Problem 14.10 (MLE from Minimum of Exponential (Final Fall 2022, Q5)).

Problem. Three batteries $X_1, X_2, X_3 \overset{\text{iid}}{\sim} \text{Exp}(\lambda)$. Let $Y_{(1)} = \min$. Given n observations of $Y_{(1)}$, find the MLE of λ.

Solution. $Y_{(1)} \sim \text{Exp}(3\lambda)$, so $f_{Y_{(1)}}(y) = 3\lambda e^{-3\lambda y}$, $y > 0$.
Given n observations $y_1, \ldots, y_n$ of $Y_{(1)}$:

$$L(\lambda) = (3\lambda)^n e^{-3\lambda \sum y_i}, \quad \ell(\lambda) = n \ln(3\lambda) - 3\lambda \sum y_i.$$

$$\ell'(\lambda) = \frac{n}{\lambda} - 3\sum y_i = 0 \implies \boxed{\hat{\lambda} = \frac{n}{3\sum Y_i} = \frac{1}{3\bar{Y}}}.$$

Appendix A

Past Tests and Exams with Full Solutions

This appendix organizes problems from past STA260 tests and exams by topic, with cross-references to the relevant chapters.

A.1 Term Test 1 Problems

Topics: CLT, Normal approximation, Confidence intervals, Sampling distributions.

Solved Problem A.1 (Binomial Normal Approximation (TT1, Fall 2024, Q1a)).

Problem. $n = 300$, $p = 0.05$. Find $\mathbb{P}(X > 20)$.
Uses the Normal approximation to the Binomial (see Chapter 2).

Solution. $\mu = np = 15$, $\sigma^2 = np(1-p) = 14.25$, $\sigma = 3.776$.
Continuity correction: $\mathbb{P}(X > 20) = \mathbb{P}(X \geq 20.5)$.

$$Z = \frac{20.5 - 15}{3.776} = 1.457.$$

$$\mathbb{P}(Z > 1.457) = 1 - \Phi(1.457) \approx 1 - 0.927 = \boxed{0.073}.$$

Solved Problem A.2 (Sample Size for CI (Test 2, Fall 2023, Q2)).

Problem. $Y \sim \mathcal{N}(\mu, 9)$. Find n such that $\mathbb{P}(\bar{Y} - 1 < \mu < \bar{Y} + 1) = 0.90$.
Uses the sample-size formula for confidence intervals (see Chapter 13).

Solution. Need $z_{0.05} \cdot \sigma/\sqrt{n} = 1$. So $1.645 \cdot 3/\sqrt{n} = 1$:

$$\sqrt{n} = 4.935, \quad n = 24.35 \implies \boxed{n \geq 25}.$$

A.2 Term Test 2 Problems

Topics: MLE, Sufficiency, Fisher Information, CRLB, Efficiency, Neyman–Pearson.

Solved Problem A.3 (Beta Distribution MLE (Test 2, Fall 2024, Q5)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} f(y;\theta) = \theta y^{\theta-1}$, $0 < y < 1$.
(a) Find $\hat{\theta}_{\text{MLE}}$.
(b) Show $\prod Y_i$ (equivalently $\sum \ln Y_i$) is sufficient.
(c) Find $\mathcal{I}(\theta)$ and the CRLB.
(d) Is $\hat{\theta}$ the MVUE?
See Chapters 9, 10, 11.

Solution.
(a) $\ell(\theta) = n \ln \theta + (\theta - 1) \sum \ln y_i$.

$$\ell'(\theta) = \frac{n}{\theta} + \sum \ln y_i = 0 \implies \boxed{\hat{\theta} = \frac{-n}{\sum \ln Y_i}}.$$

Note $\ln Y_i < 0$ since $0 < Y_i < 1$, so $\hat{\theta} > 0$.
(b) $L(\theta) = \theta^n \prod y_i^{\theta-1} = \theta^n e^{(\theta-1)\sum \ln y_i} = \underbrace{\theta^n e^{\theta \sum \ln y_i}}_{g(T,\theta)} \cdot \underbrace{e^{-\sum \ln y_i}}_{h(\boldsymbol{y})}$.
By Factorization, $T = \sum \ln Y_i$ is sufficient. ✓
(c) $\ln f = \ln \theta + (\theta - 1) \ln y$. $\frac{\partial^2}{\partial \theta^2} \ln f = -1/\theta^2$.

$$\mathcal{I}(\theta) = \frac{1}{\theta^2}, \quad \text{CRLB} = \frac{\theta^2}{n}.$$

(d) Check $\text{Var}(\hat{\theta})$: Since $-\ln Y_i \sim \text{Exp}(1/\theta)$, $T = -\sum \ln Y_i \sim \text{Gamma}(n, 1/\theta)$, so $\hat{\theta} = n/T$. The variance can be computed via the inverse Gamma distribution; one can show $\text{Var}(\hat{\theta}) = \theta^2/(n-2)$ for $n > 2$, which exceeds the CRLB. So $\hat{\theta}$ is *not* the MVUE, but it is asymptotically efficient.

Solved Problem A.4 (Consistency for Beta (Test 2, Fall 2023, Q3)).

Problem. $f(y;\theta) = \theta y^{\theta-1}$, $0 < y < 1$. Show $\bar{Y}$ is a consistent estimator of $\theta/(\theta+1)$.
See Chapter 9.

Solution. $Y \sim \text{Beta}(\theta, 1)$, so $\mathbb{E}[Y] = \theta/(\theta+1)$ and $\text{Var}(Y) = \theta/[(\theta+1)^2(\theta+2)]$.
By the Weak Law of Large Numbers: $\bar{Y} \xrightarrow{p} \mathbb{E}[Y] = \theta/(\theta+1)$ as $n \to \infty$.
Therefore $\bar{Y}$ is consistent for $\theta/(\theta+1)$. □

Solved Problem A.5 (Minimizing MSE with Constant Factor (Test 1, Fall 2022, Q1)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \sigma^2)$. Consider $T_n = cS^2$ for constant $c > 0$. Find c that minimizes $\text{MSE}(T_n)$ as an estimator of σ^2.

Solution. $\mathbb{E}[cS^2] = c\sigma^2$. Bias $= (c-1)\sigma^2$. $\mathrm{Var}(cS^2) = c^2\,\mathrm{Var}(S^2) = c^2 \cdot \frac{2\sigma^4}{n-1}$.

$$\mathrm{MSE}(cS^2) = \frac{2c^2\sigma^4}{n-1} + (c-1)^2\sigma^4.$$

Set $\frac{d}{dc}\,\mathrm{MSE} = 0$:

$$\frac{4c\sigma^4}{n-1} + 2(c-1)\sigma^4 = 0 \implies c\left(\frac{4}{n-1} + 2\right) = 2 \implies c = \frac{2}{2 + 4/(n-1)} = \frac{n-1}{n+1}.$$

$$\boxed{c^* = \frac{n-1}{n+1}}.$$

This biased estimator $\frac{n-1}{n+1}S^2$ has strictly smaller MSE than the unbiased S^2 — a classic bias–variance tradeoff.

Solved Problem A.6 (Pivotal CI for Exponential (Test 2, Fall 2023, Q6)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathrm{Exp}(\theta)$. Construct a $(1-\alpha)\%$ CI for θ using the pivotal quantity $Q = 2\sum Y_i/\theta$.

Solution. Since $Y_i \sim \mathrm{Exp}(\theta) = \mathrm{Gamma}(1, \theta)$, $\sum Y_i \sim \mathrm{Gamma}(n, \theta)$.
Then $Q = 2\sum Y_i/\theta \sim \chi^2(2n)$ (since $\mathrm{Gamma}(n,\theta) \cdot 2/\theta = \mathrm{Gamma}(n, 2) = \chi^2(2n)$).

$$\mathbb{P}\left(\chi^2_{1-\alpha/2,2n} \le \frac{2\sum Y_i}{\theta} \le \chi^2_{\alpha/2,2n}\right) = 1 - \alpha.$$

Invert:

$$\boxed{\left[\frac{2\sum Y_i}{\chi^2_{\alpha/2,2n}}, \frac{2\sum Y_i}{\chi^2_{1-\alpha/2,2n}}\right]}.$$

Solved Problem A.7 (Gamma CI via Chi-Square Pivot (Summer 2025 Test 2, Q4)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathrm{Gamma}(\alpha, \theta)$ with α known. Construct a CI for θ.

Solution. $\sum Y_i \sim \mathrm{Gamma}(n\alpha, \theta)$, so $Q = 2\sum Y_i/\theta \sim \chi^2(2n\alpha)$ (when $2n\alpha$ is an integer).

$$\boxed{\left[\frac{2\sum Y_i}{\chi^2_{\alpha/2,2n\alpha}}, \frac{2\sum Y_i}{\chi^2_{1-\alpha/2,2n\alpha}}\right]}.$$

A.3 Term Test 2 Additional Problems

Solved Problem A.8 (Two-Group Training Comparison (Test 2, Fall 2024, Q1)).

Problem. Two training methods: $n_1 = 15$, $\bar{X}_1 = 446$, $S_1 = 42$ and $n_2 = 15$, $\bar{X}_2 = 534$, $S_2 = 45$. Construct a 95% CI for $\mu_1 - \mu_2$ (equal variances assumed).

Solution. $S_p^2 = \frac{14(42)^2+14(45)^2}{28} = \frac{24696+28350}{28} = 1894.5$, $S_p = 43.53$.
$df = 28$, $t_{0.025,28} = 2.048$.

$$(446 - 534) \pm 2.048 \cdot 43.53\sqrt{\frac{2}{15}} = -88 \pm 2.048(43.53)(0.3651) = -88 \pm 32.55.$$

$$\boxed{95\% \text{ CI} : (-120.55, -55.45)}.$$

Since 0 is not in the interval, the methods produce significantly different results.

A.4 Final Exam Problems

Solved Problem A.9 (Normal $N(0, \theta^2)$ Complete Analysis (Final Fall 2023, Q6)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(0, \theta^2)$ with unknown $\theta > 0$.
(a) Find $\hat{\theta}_{\text{MLE}}$.
(b) Show sufficiency.
(c) Find $\mathcal{I}(\theta)$.
(d) Determine if the MLE is MVUE.
See Chapters 9, 10, 11.

Solution.
(a) $\ell(\theta) = -n \ln \theta - n \ln \sqrt{2\pi} - \frac{\sum y_i^2}{2\theta^2}$.

$$\ell'(\theta) = -\frac{n}{\theta} + \frac{\sum y_i^2}{\theta^3} = 0 \implies \theta^2 = \frac{\sum y_i^2}{n}.$$

$$\boxed{\hat{\theta}_{\text{MLE}} = \sqrt{\frac{1}{n}\sum Y_i^2}}.$$

(b) $L(\theta) = (2\pi\theta^2)^{-n/2} \exp\left(-\frac{\sum y_i^2}{2\theta^2}\right) = g(\sum y_i^2, \theta) \cdot 1$.
By Factorization, $T = \sum Y_i^2$ is sufficient. ✓
(c) $\ln f = -\ln \theta - \frac{1}{2}\ln(2\pi) - \frac{y^2}{2\theta^2}$.

$$\frac{\partial}{\partial\theta} \ln f = -\frac{1}{\theta} + \frac{y^2}{\theta^3}, \quad \frac{\partial^2}{\partial\theta^2} \ln f = \frac{1}{\theta^2} - \frac{3y^2}{\theta^4}.$$

$$\mathcal{I}(\theta) = -\frac{1}{\theta^2} + \frac{3\,\mathbb{E}[Y^2]}{\theta^4} = -\frac{1}{\theta^2} + \frac{3\theta^2}{\theta^4} = \boxed{\frac{2}{\theta^2}}.$$

CRLB: $\theta^2/(2n)$.
(d) Consider $\hat{\theta}^2 = \frac{1}{n}\sum Y_i^2$. Since $Y_i^2/\theta^2 \sim \chi^2(1)$, $\sum Y_i^2/\theta^2 \sim \chi^2(n)$, so $\mathbb{E}[\hat{\theta}^2] = \theta^2$ (i.e., $\hat{\theta}^2$ is unbiased for θ^2). However, $\hat{\theta} = \sqrt{\hat{\theta}^2}$ is biased for θ due to the concavity of the square root. The MVUE investigation requires further analysis.

Solved Problem A.10 (Shifted Exponential MLE (Final Fall 2023, Q5)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Exp}(\beta)$ shifted: $f(y;\theta) = \frac{1}{\beta}e^{-(y-\theta)/\beta}$, $y > \theta$.
(a) Find MoM estimator.
(b) Find MLE.
See Chapter 9.

Solution.
(a) $\mathbb{E}[Y] = \theta + \beta$. If $\beta = 1$ (given): $\bar{Y} = \theta + 1$, so $\boxed{\hat{\theta}_{\text{MoM}} = \bar{Y} - 1}$.
(b) $L(\theta) = \beta^{-n}\exp\left(-\frac{\sum(y_i-\theta)}{\beta}\right)$ subject to $\theta \le y_i$ for all i, i.e., $\theta \le Y_{(1)}$.
The log-likelihood $\ell = -n\ln\beta + n\theta/\beta - \sum y_i/\beta$ is *increasing* in θ.
Maximize by taking θ as large as possible: $\boxed{\hat{\theta}_{\text{MLE}} = Y_{(1)}}$.

Solved Problem A.11 (Bernoulli CRLB and MVUE (Final Fall 2025, Q5)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Bernoulli}(p)$.
(a) Compute the CRLB for p.
(b) Show $\hat{p} = \bar{Y}$ is the MVUE.
(c) Find the MVUE of $p(1-p)$.

Solution.
(a) $\mathcal{I}(p) = 1/[p(1-p)]$ (computed in Chapter 11). So CRLB $= \frac{p(1-p)}{n}$.
(b) $\text{Var}(\bar{Y}) = p(1-p)/n = \text{CRLB}$. Since the MLE achieves the CRLB, $\bar{Y}$ is **efficient** and therefore the MVUE.
(c) By Rao–Blackwell (see Chapter 11):

$$\boxed{\text{MVUE of } p(1-p) = \frac{T(n-T)}{n(n-1)}, \quad T = \sum Y_i}.$$

Solved Problem A.12 (Pareto MLE and MoM (Final Fall 2025, Q4)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Pareto}(\theta)$, $f(y;\theta) = \theta/y^{\theta+1}$, $y > 1$.
(a) Find $\hat{\theta}_{\text{MLE}}$. (b) Find $\hat{\theta}_{\text{MoM}}$.
See Chapter 9.

Solution.

(a) $\ell(\theta) = n \ln \theta - (\theta + 1) \sum \ln y_i$. $\ell'(\theta) = n/\theta - \sum \ln y_i = 0$:

$$\boxed{\hat{\theta}_{\text{MLE}} = \frac{n}{\sum \ln Y_i}}.$$

(b) $\mathbb{E}[Y] = \theta/(\theta - 1)$ for $\theta > 1$. Set $\bar{Y} = \theta/(\theta - 1)$:

$$\boxed{\hat{\theta}_{\text{MoM}} = \frac{\bar{Y}}{\bar{Y} - 1}}.$$

Solved Problem A.13 (Beta Sufficiency and MVUE (Final Fall 2025, Q3)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Beta}(\theta, 1)$, $f(y; \theta) = \theta y^{\theta - 1}$.
(a) Show $T = \prod Y_i$ is sufficient. (b) Show it belongs to the exponential family. (c) Find the MVUE of θ.
See Chapters 10 and 11.

Solution.
(a) By Factorization: $L(\theta) = \theta^n e^{\theta \sum \ln y_i} \cdot e^{-\sum \ln y_i}$. So $T = \sum \ln Y_i$ (equivalently $\prod Y_i$) is sufficient. ✓
(b) $f(y; \theta) = \theta \cdot 1 \cdot \exp[\theta \cdot \ln y + \ln 1]$. This is the exponential family form $a(\theta)b(y)\exp[c(\theta)d(y)]$ with $a(\theta) = \theta$, $c(\theta) = \theta$, $d(y) = \ln y$.
By exponential family theory, $T = \sum d(Y_i) = \sum \ln Y_i$ is complete sufficient.
(c) Since $-\ln Y_i \sim \text{Exp}(1/\theta)$, $-T \sim \text{Gamma}(n, 1/\theta)$, and $\mathbb{E}[n/(-T)] = n \cdot \theta/(n-1)$.
So $\mathbb{E}[(n-1)/(-T)] = \theta$:

$$\boxed{\hat{\theta}_{\text{MVUE}} = \frac{-(n-1)}{\sum \ln Y_i}}.$$

Solved Problem A.14 (Binomial Normal Approx. (Final Fall 2024, Q1)).

Problem. $Y \sim \text{Bin}(400, 0.2)$. Find $\mathbb{P}(Y < 60)$ using normal approximation with continuity correction.

Solution. $\mu = 400(0.2) = 80$, $\sigma^2 = 400(0.2)(0.8) = 64$, $\sigma = 8$.
Check: $np = 80 \geq 10$ ✓, $n(1-p) = 320 \geq 10$ ✓.
Continuity correction: $\mathbb{P}(Y < 60) = \mathbb{P}(Y \leq 59) \approx \mathbb{P}(Y \leq 59.5)$.

$$Z = \frac{59.5 - 80}{8} = -2.5625, \quad \mathbb{P}(Z \leq -2.5625) \approx \boxed{0.0052}.$$

Solved Problem A.15 (Left-Handed Proportion (Final Fall 2023, Q1)).

Problem. 10% of population is left-handed. In a class of $n = 150$, find $\mathbb{P}(15 \leq X \leq 20)$.

Solution. $X \sim \text{Bin}(150, 0.10)$. $\mu = 15$, $\sigma^2 = 150(0.1)(0.9) = 13.5$, $\sigma = 3.674$.
Continuity correction: $\mathbb{P}(14.5 \le X \le 20.5)$.

$$Z_L = \frac{14.5 - 15}{3.674} = -0.136, \quad Z_U = \frac{20.5 - 15}{3.674} = 1.497.$$

$$\mathbb{P}(-0.136 \le Z \le 1.497) = \Phi(1.497) - \Phi(-0.136) \approx 0.9327 - 0.4459 = \boxed{0.487}.$$

Solved Problem A.16 (Beta MLE, Sufficiency, CRLB, Efficiency (Final Fall 2024, Q5)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Beta}(\theta, 1)$, $f(y; \theta) = \theta y^{\theta - 1}$.
(a) Show $\hat{\theta}_{\text{MLE}} = -n / \sum \ln Y_i$. (b) Sufficiency of $\sum \ln Y_i$. (c) Find CRLB. (d) Is $\frac{n-1}{n}\hat{\theta}$ efficient?

Solution.
(a)–(b) See previous problems. $\hat{\theta}_{\text{MLE}} = -n / \sum \ln Y_i$, $T = \sum \ln Y_i$ is sufficient.
(c) $\mathcal{I}(\theta) = 1/\theta^2 \implies \text{CRLB} = \theta^2/n$.
(d) Consider $\hat{\theta}^* = \frac{n-1}{n}\hat{\theta}_{\text{MLE}} = \frac{-(n-1)}{\sum \ln Y_i}$, which is unbiased (shown earlier).
$\text{Var}(\hat{\theta}^*) = \theta^2/(n-2)$ for $n > 2$ (via inverse Gamma properties).
$\text{eff} = \frac{\text{CRLB}}{\text{Var}(\hat{\theta}^*)} = \frac{\theta^2/n}{\theta^2/(n-2)} = \frac{n-2}{n} < 1$.
So $\hat{\theta}^*$ is **not efficient** (it does not achieve the CRLB), though its efficiency $\to 1$ as $n \to \infty$.

Solved Problem A.17 (NP Test for Normal with Power (Final Fall 2025, Q6)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, 4)$. Test $H_0 : \mu = 10$ vs $H_A : \mu = 12$, $n = 16$, $\alpha = 0.05$.
(a) Rejection region. (b) Critical value. (c) Power.
See Chapter 12.

Solution.
(a) RR : $\{\bar{Y} \ge c\}$ (since $\mu_1 = 12 > \mu_0 = 10$).
(b) Under H_0: $\bar{Y} \sim \mathcal{N}(10, 4/16) = \mathcal{N}(10, 0.25)$. $c = 10 + z_{0.05} \cdot 0.5 = 10 + 1.645(0.5) = \boxed{10.823}$.
(c) Under H_A: $\bar{Y} \sim \mathcal{N}(12, 0.25)$.

$$\text{Power} = \mathbb{P}(\bar{Y} \ge 10.823 \mid \mu = 12) = \mathbb{P}\left(Z \ge \frac{10.823 - 12}{0.5}\right) = \mathbb{P}(Z \ge -2.355) \approx \boxed{0.991}.$$

A.5 Additional Estimation and Efficiency Problems

Solved Problem A.18 (Poisson MVUE and Delta Method (Test 2, Fall 2021, Q1)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Poisson}(\lambda)$.
(a) Show $\bar{Y}$ is the MVUE of λ.
(b) Find an estimator of $e^{-\lambda}$ and its asymptotic distribution.

Solution.
(a) $\bar{Y}$ is unbiased: $\mathbb{E}[\bar{Y}] = \lambda$. $\text{Var}(\bar{Y}) = \lambda/n$.
$\mathcal{I}(\lambda) = 1/\lambda$, so CRLB $= \lambda/n$. Since $\text{Var}(\bar{Y}) = \text{CRLB}$, $\bar{Y}$ is efficient and hence the MVUE. ✓
(b) By invariance of MLEs: $\widehat{e^{-\lambda}} = e^{-\bar{Y}}$.
By Delta Method with $g(\lambda) = e^{-\lambda}$, $g'(\lambda) = -e^{-\lambda}$:

$$\sqrt{n}(e^{-\bar{Y}} - e^{-\lambda}) \xrightarrow{d} \mathcal{N}\Big(0,\ [g'(\lambda)]^2 \cdot \lambda\Big) = \mathcal{N}(0, \lambda e^{-2\lambda}).$$

$$\boxed{e^{-\bar{Y}} \dot{\sim} \mathcal{N}\left(e^{-\lambda},\ \frac{\lambda e^{-2\lambda}}{n}\right)}.$$

Solved Problem A.19 (Uniform Estimator Efficiency (Test 2, Fall 2021, Q2)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Uniform}(\theta, \theta+1)$. $\hat{\theta}_1 = \bar{Y} - 1/2$, $\hat{\theta}_2 = Y_{(n)} - n/(n+1)$. Find $\text{eff}(\hat{\theta}_1, \hat{\theta}_2)$.

Solution. $\text{Var}(\hat{\theta}_1) = \text{Var}(\bar{Y}) = 1/(12n)$.
For $Y_{(n)}$: $f_{Y_{(n)}}(y) = n(y-\theta)^{n-1}$ on $[\theta, \theta+1]$. $\mathbb{E}[Y_{(n)}] = \theta + n/(n+1)$, $\text{Var}(Y_{(n)}) = n/[(n+1)^2(n+2)]$.
$\text{Var}(\hat{\theta}_2) = \text{Var}(Y_{(n)}) = n/[(n+1)^2(n+2)]$.

$$\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)} = \frac{n/[(n+1)^2(n+2)]}{1/(12n)} = \frac{12n^2}{(n+1)^2(n+2)}.$$

For large n, eff $\approx 12/n \to 0$: $\hat{\theta}_2$ is *much* better than $\hat{\theta}_1$. $\boxed{\text{eff}(\hat{\theta}_1, \hat{\theta}_2) = \frac{12n^2}{(n+1)^2(n+2)}}$.

Solved Problem A.20 (Pareto Distribution Full Analysis (Final Fall 2022, Q4)).

Problem. $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Pareto}(\theta)$: $f(x;\theta) = (\theta-1)x^{-\theta}$, $x \geq 1$, $\theta > 2$.
(a) Find MLE. (b) Sufficiency. (c) Fisher information. (d) Approximate CI.

Solution.
(a) $\ell(\theta) = n\ln(\theta-1) - \theta\sum \ln x_i$.

$$\ell'(\theta) = \frac{n}{\theta-1} - \sum \ln x_i = 0 \implies \boxed{\hat{\theta} = 1 + \frac{n}{\sum \ln X_i}}.$$

(b) By Factorization: $L(\theta) = (\theta-1)^n \exp(-\theta\sum \ln x_i) = g(\sum \ln x_i, \theta)\cdot 1$.
$T = \sum \ln X_i$ is sufficient. ✓
(c) $\frac{\partial^2}{\partial\theta^2}\ln f = -\frac{1}{(\theta-1)^2}$, so $\mathcal{I}(\theta) = \frac{1}{(\theta-1)^2}$.
(d) With $n = 25$, $\sum \ln x_i = 5$: $\hat{\theta} = 1 + 5 = 6$.
Asymptotic variance: $\mathrm{Var}(\hat{\theta}) \approx 1/[n\,\mathcal{I}(\theta)] = (\theta-1)^2/n$. Estimated SE $= (\hat{\theta}-1)/\sqrt{n} = 5/5 = 1$.
Approximate 90% CI: $6 \pm 1.645(1) = \boxed{(4.355, 7.645)}$.

Solved Problem A.21 (Combined Unbiased Estimator — Variance Minimization (Final Fall 2022, Q1)).

Problem. Three estimators X_1, X_2, X_3 with $\mathbb{E}[X_i] = \theta$ for $i = 1, 3$ and $\mathbb{E}[X_2] = 2\theta - 1$. Consider $\hat{\theta} = (\alpha/2)X_1 + (\alpha-1)X_2 + (\alpha/2)X_3$.
(a) Find α for unbiasedness.
(b) Find α that minimizes $\mathrm{Var}(\hat{\theta})$.

Solution.
(a) $\mathbb{E}[\hat{\theta}] = (\alpha/2)\theta + (\alpha-1)(2\theta-1) + (\alpha/2)\theta = \alpha\theta + 2\alpha\theta - 2\theta - \alpha + 1 = (3\alpha-2)\theta + (1-\alpha)$.
For unbiasedness: $(3\alpha-2)\theta + (1-\alpha) = \theta$ for all θ. Coefficients: $3\alpha - 2 = 1 \implies \alpha = 1$ and $1 - \alpha = 0 \implies \alpha = 1$. ✓
But let's check more carefully — since the problem asks to minimize variance subject to unbiasedness, if $\alpha = 1$: $\hat{\theta} = X_1/2 + 0\cdot X_2 + X_3/2 = (X_1 + X_3)/2$.
(b) $\mathrm{Var}(\hat{\theta}) = (\alpha/2)^2\,\mathrm{Var}(X_1) + (\alpha-1)^2\,\mathrm{Var}(X_2) + (\alpha/2)^2\,\mathrm{Var}(X_3)$.
With given variances: $\mathrm{Var}(\hat{\theta}) = 5\alpha^2 - 8\alpha + 4$.
$\frac{d}{d\alpha}\mathrm{Var} = 10\alpha - 8 = 0 \implies \boxed{\alpha = 0.8}$.

Solved Problem A.22 (Uniform MoM vs MLE — MSE Comparison (Final Summer 2025, Q2)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathrm{Uniform}(0, \theta)$.
(a) MoM estimator. (b) MLE. (c) Compare MSE.

Solution.
(a) $\mathbb{E}[Y] = \theta/2$, so $\bar{Y} = \theta/2 \implies \boxed{\hat{\theta}_{\text{MoM}} = 2\bar{Y}}$.
(b) $L(\theta) = \theta^{-n}$ for $\theta \geq \max(y_i) = y_{(n)}$. Decreasing in θ, so $\boxed{\hat{\theta}_{\text{MLE}} = Y_{(n)}}$.
(c) MoM: $\mathrm{Var}(2\bar{Y}) = 4\,\mathrm{Var}(\bar{Y}) = 4\cdot\frac{\theta^2/12}{n} = \frac{\theta^2}{3n}$. Unbiased, so MSE $= \theta^2/(3n)$.

MLE: $\mathbb{E}[Y_{(n)}] = n\theta/(n+1) \neq \theta$ (biased). Bias $= -\theta/(n+1)$. $\mathrm{Var}(Y_{(n)}) = n\theta^2/[(n+1)^2(n+2)]$.

$$\mathrm{MSE}(Y_{(n)}) = \frac{n\theta^2}{(n+1)^2(n+2)} + \frac{\theta^2}{(n+1)^2} = \frac{2\theta^2}{(n+1)(n+2)}.$$

Since $\frac{2}{(n+1)(n+2)} < \frac{1}{3n}$ for $n \geq 1$, the **MLE has smaller MSE** despite being biased.

Solved Problem A.23 (Rayleigh MVUE (Final Summer 2025, Q3)).

Problem. $Y \sim \mathrm{Rayleigh}(\theta)$: $f(y;\theta) = \frac{2y}{\theta^2} e^{-y^2/\theta^2}$, $y > 0$.
(a) Show it's an exponential family and find sufficient statistic.
(b) Find the MVUE of θ^2.

Solution.
(a) $f(y;\theta) = \underbrace{2y}_{b(y)} \cdot \underbrace{\theta^{-2}}_{a(\theta)} \cdot \exp\left[\underbrace{(-1/\theta^2)}_{c(\theta)} \cdot \underbrace{y^2}_{d(y)}\right]$.
Exponential family. $T = \sum Y_i^2$ is complete sufficient.
(b) $\mathbb{E}[Y^2] = \theta^2$ (from the Rayleigh distribution). So $\mathbb{E}[\sum Y_i^2/n] = \theta^2$.
$\boxed{\hat{\theta}^2_{\mathrm{MVUE}} = \frac{1}{n}\sum Y_i^2}$ is unbiased for θ^2 and is a function of the complete sufficient statistic.
By Lehmann–Scheffé, it is the MVUE.

Solved Problem A.24 (Bernoulli Rao–Blackwell for $(1-p)^2$ (Final Summer 2025, Q4)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \mathrm{Bernoulli}(p)$. Find the MVUE of $(1-p)^2$.

Solution. Unbiased estimator: $W = (1-Y_1)(1-Y_2)$. Note $\mathbb{E}[W] = (1-p)^2$. ✓
$T = \sum Y_i$ is complete sufficient. Apply Rao–Blackwell:

$$\theta^*(t) = \mathbb{E}[W \mid T = t] = \mathbb{P}(Y_1 = 0, Y_2 = 0 \mid T = t).$$

Given $T = t$: the remaining $n-2$ observations contribute t successes (if $Y_1 = Y_2 = 0$).

$$\theta^*(t) = \frac{\binom{n-2}{t}}{\binom{n}{t}} = \frac{(n-t)(n-t-1)}{n(n-1)}.$$

$$\boxed{\text{MVUE of } (1-p)^2 = \frac{(n-T)(n-T-1)}{n(n-1)}, \quad T = \sum Y_i}.$$

Solved Problem A.25 (Exponential MLE and Efficiency (Final Summer 2025, Q5)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$ with rate θ: $f(y;\theta) = \theta e^{-\theta y}$.
(a) Find MLE. (b) Fisher information. (c) Efficiency of $\hat{\theta}^* = \frac{n-1}{n}\hat{\theta}_{\text{MLE}}$.

Solution.
(a) $\ell(\theta) = n \ln \theta - \theta \sum y_i$. $\ell'(\theta) = n/\theta - \sum y_i = 0$:

$$\boxed{\hat{\theta}_{\text{MLE}} = \frac{n}{\sum Y_i} = \frac{1}{\bar{Y}}}.$$

(b) $\frac{\partial^2}{\partial \theta^2} \ln f = -1/\theta^2$, so $\mathcal{I}(\theta) = 1/\theta^2$, $I_n(\theta) = n/\theta^2$.
CRLB $= \theta^2/n$.
(c) $\hat{\theta}^*$ is constructed to be unbiased (from the Gamma distribution of $\sum Y_i$).
$\text{Var}(\hat{\theta}^*) = \left(\frac{n-1}{n}\right)^2 \text{Var}(\hat{\theta}_{\text{MLE}})$. Since $\sum Y_i \sim \text{Gamma}(n, 1/\theta)$, $\hat{\theta}_{\text{MLE}} = n/\sum Y_i$ has $\text{Var}(\hat{\theta}_{\text{MLE}}) = \theta^2/(n-2)$ for $n > 2$.

$$\text{Var}(\hat{\theta}^*) = \frac{(n-1)^2}{n^2} \cdot \frac{\theta^2}{n-2}.$$

Efficiency: $\text{eff} = \frac{\text{CRLB}}{\text{Var}(\hat{\theta}^*)} = \frac{\theta^2/n}{\frac{(n-1)^2\theta^2}{n^2(n-2)}} = \frac{n(n-2)}{(n-1)^2} < 1$.

A.6 Final Exam Fall 2025 Version B

Solved Problem A.26 (Uniform$(0,1)$ Range and Midpoint (Final Fall 2025 B, Q1)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Uniform}(0,1)$. Define $R = Y_{(n)} - Y_{(1)}$ (range) and $M = (Y_{(1)} + Y_{(n)})/2$ (midpoint).
(a) Find the joint PDF $f_{R,M}(r,m)$.
(b) Find $f_R(r)$ and $\mathbb{E}[R]$.

Solution.
(a) Joint PDF of $(Y_{(1)}, Y_{(n)})$ for Uniform$(0,1)$:

$$f_{Y_{(1)},Y_{(n)}}(y_1, y_n) = n(n-1)(y_n - y_1)^{n-2}, \quad 0 < y_1 < y_n < 1.$$

Transform: $R = Y_{(n)} - Y_{(1)}$, $M = (Y_{(1)} + Y_{(n)})/2$, Jacobian $|J| = 1$.
Support: $0 < r < 1$, $r/2 < m < 1 - r/2$.

$$\boxed{f_{R,M}(r,m) = n(n-1)r^{n-2}}.$$

(b) Integrate over m:

$$f_R(r) = n(n-1)r^{n-2}\int_{r/2}^{1-r/2} \mathrm{d}m = \boxed{n(n-1)r^{n-2}(1-r)}, \quad 0 < r < 1.$$

$$\mathbb{E}[R] = n(n-1)\int_0^1 r^{n-1}(1-r)\,\mathrm{d}r = n(n-1)\left[\frac{1}{n} - \frac{1}{n+1}\right] = \boxed{\frac{n-1}{n+1}}.$$

Solved Problem A.27 (Exponential: Comparing $\bar{Y}$ and $nY_{(1)}$ (Final Fall 2025 B, Q2)).

Problem. $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} \text{Exp}(\theta)$. Compare $\hat{\theta}_1 = \bar{Y}$ and $\hat{\theta}_2 = nY_{(1)}$.
(a) Show $\hat{\theta}_1$ is unbiased. Is $Y_{(1)}$ unbiased?
(b) Compute MSE of both.
(c) Which is preferable?

Solution.
(a) $\mathbb{E}[\bar{Y}] = \theta$ ✓. $Y_{(1)} \sim \text{Exp}(\theta/n)$, so $\mathbb{E}[Y_{(1)}] = \theta/n \neq \theta$. Biased.
For unbiasedness: $\mathbb{E}[nY_{(1)}] = n \cdot \theta/n = \theta$. ✓
(b) $\text{MSE}(\hat{\theta}_1) = \text{Var}(\bar{Y}) = \theta^2/n$.
$\text{Var}(Y_{(1)}) = \theta^2/n^2$, so $\text{Var}(nY_{(1)}) = n^2 \cdot \theta^2/n^2 = \theta^2$.
$\text{MSE}(nY_{(1)}) = \theta^2$.
(c) For $n > 1$: $\theta^2/n < \theta^2$, so $\boxed{\hat{\theta}_1 = \bar{Y} \text{ is preferable}}$. The sample mean uses information from *all* observations, while $nY_{(1)}$ only uses the minimum.

Solved Problem A.28 (Optimal Weighted Estimator with Covariance (Note Jul 14)).

Problem. Two unbiased estimators $\hat{\theta}_1, \hat{\theta}_2$ with $\text{Var}(\hat{\theta}_1) = \sigma_1^2$, $\text{Var}(\hat{\theta}_2) = \sigma_2^2$, $\text{Cov}(\hat{\theta}_1, \hat{\theta}_2) = c$. For $\hat{\theta}_3 = \alpha\hat{\theta}_1 + (1-\alpha)\hat{\theta}_2$, find the optimal α.

Solution. $\hat{\theta}_3$ is unbiased for any α: $\mathbb{E}[\hat{\theta}_3] = \alpha\theta + (1-\alpha)\theta = \theta$.
$\text{Var}(\hat{\theta}_3) = \alpha^2\sigma_1^2 + (1-\alpha)^2\sigma_2^2 + 2\alpha(1-\alpha)c$.
$\frac{d}{d\alpha}\text{Var} = 2\alpha\sigma_1^2 - 2(1-\alpha)\sigma_2^2 + 2(1-2\alpha)c = 0$.

$$\alpha(\sigma_1^2 + \sigma_2^2 - 2c) = \sigma_2^2 - c.$$

$$\boxed{\alpha^* = \frac{\sigma_2^2 - c}{\sigma_1^2 + \sigma_2^2 - 2c}}.$$

When $c = 0$ (independent): $\alpha^* = \sigma_2^2/(\sigma_1^2 + \sigma_2^2)$ (weights inversely proportional to variances).

Solved Problem A.29 (Large-Sample CI for Difference in Proportions).

Problem. Two independent samples: $\hat{p}_1 = X_1/n_1$ and $\hat{p}_2 = X_2/n_2$. Construct a $(1-\alpha)100\%$ CI for $p_1 - p_2$.

Solution. By CLT, $\hat{p}_i \dot{\sim} \mathcal{N}(p_i, p_i(1-p_i)/n_i)$. By independence:

$$\hat{p}_1 - \hat{p}_2 \dot{\sim} \mathcal{N}\left(p_1 - p_2,\ \frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}\right).$$

$$\boxed{(\hat{p}_1 - \hat{p}_2) \pm z_{\alpha/2}\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}}.$$